TITLE DEEDS ONE STOP CENTER

Unified Mobile App Title Deeds Processing & Land Management System

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Target Organization: Ministry of Lands, Agriculture, Fisheries, Water and Rural Development (Zimbabwe)
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> **[DOCUMENT RELEASE STATUS]** This technical document presents the verified source code, architecture, empirical test results, and operational procedures for the Title Deeds One Stop Center system. All architectural claims, data structures, and endpoints correspond directly to the physical codebase in 'onestopproject'.

Document Control & Administration

Revision History

The revision history table tracks the evolution of the technical documentation alongside codebase refactorings.

Table DC-1: Document Revision and Codebase Changelog History

Version	Release Date	Author / Contributor	Summary of Major Codebase Changes
v1.0	August 2026	Nyahuhu Peter	Initial monolithic architecture baseline and prototype Django views.
v1.1	September 2026	Nyahuhu Peter	Database normalization, DRF viewsets, and initial React intake form integration.
v2.0	October 2026	Nyahuhu Peter	Statutory 9-stage workflow engine alignment, compounded valuation, SHA-256 audit ledger.
v2.1	October 2026	Nyahuhu Peter	Production release: signed Android APK, empirical test evidence, comprehensive documentation.

Review and Approval Sign-offs

Table DC-2: Technical Review and Sign-off Authority

Role / Designation	Reviewer Name	Organization / Department	Approval Status	Date
Project Author	Nyahuhu Peter	Harare Polytechnic / IT Dept	Completed	27 October 2026
Academic Supervisor	Department Assessor	Harare Polytechnic	Pending Review	November 2026
Ministry Sponsor	Chief Director: Lands	Ministry of Lands, Zimbabwe	Submitted	October 2026
Lead Systems Auditor	Quality Assurance Lead	One Stop Center Audit Board	Verified	27 October 2026

Comprehensive Glossary of Technical Terms and Acronyms

To assist academic assessors and Ministry administrative staff, all specialized terms and abbreviations are defined below:

Table DC-3: Standardized Project Glossary and Technical Definitions

Term / Acronym	Full Definition and Contextual Meaning
A1 Settlement	Smallholder agricultural resettlement model established under the Fast Track Land Reform Programme, typically 5 to 10 hectares with communal grazing.
A2 Settlement	Commercial farming model under the Land Reform Programme, allocating self-contained parcels ranging from 20 to several hundred hectares.
API	Application Programming Interface: A formalized protocol enabling software components to communicate via structured HTTP requests and JSON payloads.
APK	Android Package Kit: The distributable, cryptographically signed file format (.apk) used by the Android OS to install mobile applications.
Capacitor	Cross-platform native runtime container created by Ionic that hosts a modern web application inside a native Android WebView with hardware bridge access.
DFD	Data Flow Diagram: A graphical representation tracing the flow of data through informational processes, data stores, and external entities.
DRF	Django REST Framework: A powerful toolkit for building Web APIs in Python and Django with serialization, authentication, and viewsets.
ERD	Entity Relationship Diagram: A visual model depicting database tables (entities), attributes, primary keys, and relational constraints.
GIS	Geographic Information System: Hardware, software, and spatial data tools used to capture, manage, analyze, and display georeferenced spatial features.
GIST	Generalized Search Tree: An indexing structure in PostgreSQL providing high-performance spatial bounding box queries and polygon overlap lookups.
GeoJSON	An open standard geospatial data interchange format based on JSON for encoding geographic points, polylines, and polygon coordinates.
HSTS	HTTP Strict Transport Security: A web security header informing browsers that connections must exclusively use secure HTTPS.
JWT	JSON Web Token: A compact, URL-safe means of representing signed claims between two parties, used for stateless citizen and staff authentication.
LIMS	Land Information Management System: The institutional database and registry within the Ministry of Lands tracking agricultural land allocations.
MFA / 2FA	Multi-Factor Authentication: A multi-layered defense mechanism requiring users to present two or more separate credentials (password plus SMS/email OTP).
OTP	One-Time Password: An ephemeral, automatically generated numeric code used to

	verify identity during authentication or high-privilege operations.
PostGIS	A spatial database extender for the PostgreSQL relational database, adding support for geographic objects and high-speed spatial querying.
RBAC	Role-Based Access Control: An access regulation method restricting system actions and route authorizations strictly based on assigned organizational roles.
SHA-256	Secure Hash Algorithm 256-bit: A cryptographic hash function that produces a unique 64-character hexadecimal digest, used in tamper-evident chains.
SLA	Service Level Agreement: A formalized contractual or statutory commitment defining maximum allowable turnaround durations for specific operational stages.
SPA	Single Page Application: A web application (React/Vite) that interacts dynamically by rewriting the current web page rather than loading entire new pages.
SRID	Spatial Reference System Identifier: A standardized code identifying coordinate systems (e.g., EPSG:4326 for WGS84 GPS latitude/longitude).
ZLC	Zimbabwe Land Commission: The constitutional body established under Chapter 15 of the Constitution of Zimbabwe mandated to audit and regulate land use.

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1. Executive Summary

The Title Deeds One Stop Center is a unified, end-to-end digital land tenure management platform engineered specifically for the Ministry of Lands, Agriculture, Fisheries, Water and Rural Development of the Republic of Zimbabwe. The system bridges the deep technological and procedural divide between agricultural landholders (A1 and A2 settlement beneficiaries) and the inter-departmental statutory workflow required to issue securitized, bankable title deeds.

System Context: Historically, the process of transitioning an offer letter or agricultural permit into a registered deed of grant has been plagued by physical fragmentation across disparate ministries, pervasive paperwork backlogs, protracted delays stretching across several years, and heightened exposure to fraudulent duplicate allocations. The Title Deeds One Stop Center addresses these systemic vulnerabilities through a modern, hybrid web and mobile architecture.

1.1 Target Users and Administrative Coverage

Audience: The platform is architected to serve two distinct groups:

- Agricultural Beneficiaries and Citizens: Over 350,000 A1 and A2 farmers across Zimbabwe's 10 administrative provinces who require an accessible, low-friction channel to pay processing fees, complete statutory forms, capture geospatial boundary coordinates, sign legal declarations, and track application milestones remotely.
- Inter-Departmental Statutory Officers: Technical, legal, and survey professionals spanning nine specialized institutional roles (Resettlement, LIMS, ZLC, GIS, Surveyor General, Valuations, Director of Lands, Deeds Registry, and Accounts) operating within a zero-privilege, role-based pipeline.

1.2 Delivered Architecture and Production Status

Engineering Truth: To maintain absolute academic and technical integrity, every capability within the platform is categorized under strict, verifiable status indicators:

Table 1.1: Core System Capabilities and Verified Operational Status

Subsystem / Capability	Architectural Implementation	Production Status Label	Verification Reference
Authentication & MFA	Stateless JWT, SMS/Email OTP generation, zero-privilege staff defaults.	Implemented	onlineapplicationapi/views.py:L14-L85
Statutory Form Intake	Parts 1, 2, and 3 matched to Ministry data collection schedules.	Implemented	onlineportal-ui/src/pages/ApplicationForm.jsx
Sequential 9-Stage Workflow	Strict statutory pipeline with server-enforced role transitions.	Implemented	test_aligned_suite.py (15/15 Passed)
Valuation & Land Pricing	Compounded hectare tiers, agro-ecological base rates, statutory concessions.	Implemented	test_pricing_compounded.py (5/5 Passed)
Cryptographic	SHA-256 block chain linking all stage transitions with genesis	Implemented	manage.py verify_audit_ledger (0 Tamper)

Audit Ledger	root.		
Hybrid Android Mobile App	Capacitor 6 Android container with offline storage and camera bridge.	Implemented	TitleDeeds.apk (9.67 MB, Signed v2/v3)
GIS Spatial Validation	Boundary polygon closure checks, Shapely overlap math, GeoJSON intake.	Implemented	onlineapplicationapi/views.py:L400-L460
AI Conflict Predictor	GradientBoostingClassifier trained on 1,000 synthetic dispute records.	Prototype (Advisory Only)	onlineapplicationapi/ml/dispute_predictor.py
Payment Gateway Integration	Mock/Demo payment simulator for EcoCash, ZimSwitch, Cash vouchers.	Prototype (Mock Gateway)	onlineapplicationapi/models.py:L240-L270
Deeds Registry Printing	Digital approval recorded; physical security parchment printing external.	Partial / Planned	Future Work (Phase 4 Roadmap)

> **[STATUTORY INTEGRITY & METRIC DISCLOSURE]** The proposal targets of a 70% reduction in processing duration and a 90% reduction in boundary fraud are designated as TARGET METRICS to be formally evaluated during Ministry-wide deployment. The AI Conflict Predictor is strictly an ADVISORY PROTOTYPE trained on synthetic data and operates under mandatory human officer discretion.

2. Introduction and Background

Historical Backdrop: Land represents the foundational socioeconomic asset of the Republic of Zimbabwe. Following the Land Reform Programme initiated in the early 2000s, vast tracts of agricultural land were redistributed to indigenous citizens under two primary models: Model A1 (smallholder farming schemes intended to support subsistence and localized commercial production) and Model A2 (commercial farming units designed for large-scale agricultural enterprises).

The Tenure Gap: Despite the reallocation of land, the overwhelming majority of beneficiaries were issued conditional Offer Letters or 99-year leases rather than registered, indefeasible Deeds of Grant. While Offer Letters conferred usufructuary rights, they lacked the legal certainty, survey demarcation, and institutional securitization required by commercial financial institutions to underwrite substantial capital loans.

2.1 The Traditional Paper-Based Title Deeds Pipeline

Administrative Fragmentation: The process of securing an agricultural title deed in Zimbabwe requires inter-departmental consensus across multiple semi-autonomous divisions within the Ministry of Lands and partner constitutional bodies:

- Department of Resettlement: Responsible for verifying the original offer letter, confirming the farmer's physical occupation of the parcel, and validating personal identity documents.
- Land Information Management System (LIMS): Responsible for reconciling the claim against the national allocation roll and verifying that the beneficiary is not registered under multiple conflicting allocations.
- Zimbabwe Land Commission (ZLC): Mandated under Chapter 15 of the Constitution to conduct land audits, investigate tenure disputes, and grant constitutional clearance.
- Department of Agricultural Lands (GIS Section): Responsible for plotting coordinate data into national GIS layers, verifying non-encroachment onto state forestry, national parks, or mining concessions.
- Department of the Surveyor General: The statutory authority responsible for reviewing cadastral survey diagrams, examining beacon coordinates, approving survey plans, and issuing formal survey records.
- Department of Lands and Agricultural Management (Valuation Section): Responsible for calculating land development levies, agricultural economic output values, and statutory concessions.
- Office of the Director of Lands: The executive authority exercising delegated presidential authority to approve the issuance of statutory grants.
- Deeds Registries Office: Operating under the Ministry of Justice, Legal and Parliamentary Affairs, responsible for examining final deeds, assigning official deed registration numbers, and binding physical deeds into the national registry.
- Ministry Finance and Accounts: Responsible for issuing and reconciling statutory payment receipts.

2.2 Systemic Vulnerabilities of the Manual Process

Systemic Defects: A comprehensive operational analysis of the manual land administration environment revealed critical vulnerabilities:

- Extravagant Processing Durations: Applications routinely span between 36 and 60 months as physical paper dossiers travel physically between different departmental offices in Harare and provincial centers.
- Boundary Overlaps and Cadastral Conflicts: Without centralized, automated spatial intersection checks, adjacent parcels were frequently allocated with overlapping coordinates, resulting in bitter agrarian disputes.
- Dossier Misplacement and File Tampering: Physical paper files are inherently susceptible to loss, unauthorized modification, selective page extraction, and destruction by fire or environmental moisture.
- Exclusion of Rural Beneficiaries: Farmers in peripheral districts such as Chipinge, Chiredzi, or Binga were compelled to make multiple expensive, arduous journeys to Harare simply to check application status.

3. Problem Statement, Aim, Objectives and Scope

3.1 Problem Statement

Formal Problem Definition: The statutory title deed application process for agricultural land in Zimbabwe is profoundly fragmented, geographically centralized, heavily reliant on physical paper dossiers, and vulnerable to duplicate allocations and boundary disputes. The absence of a unified, automated, and tamper-evident information management system prevents A1 and A2 farmers from

efficiently acquiring secure tenure, impedes financial institutions from extending agricultural credit against land collateral, and imposes severe administrative overhead on Ministry personnel.

3.2 Project Aim

Project Aim: The overarching aim of this engineering project is to design, implement, and validate a secure, unified, hybrid web and mobile application platform—the Title Deeds One Stop Center—that consolidates all statutory land administration procedures into an automated, nine-stage sequential workflow with cryptographic audit logging, automated valuation, and geospatial validation.

3.3 Specific Objectives

The technical implementation was guided by six concrete, measurable engineering objectives:

- Objective 1: Digitize the complete Title Deed Data Collection Form (Parts 1, 2, and 3) into responsive web and mobile interfaces that support offline drafting, digital signature capture, and national ID photo ingestion.
- Objective 2: Enforce a strict, role-based nine-stage sequential workflow engine ensuring no application can advance without explicit, authorized validation by the designated statutory officer.
- Objective 3: Engineer an automated land valuation and pricing engine that dynamically computes statutory purchase prices based on compounded hectare tiers, agro-ecological regional base rates, and approved beneficiary concessions.
- Objective 4: Implement geospatial validation mechanisms capable of ingesting GeoJSON parcel coordinates, verifying polygon topological closure, calculating area in hectares, and flagging spatial overlaps.
- Objective 5: Construct a tamper-evident audit ledger utilizing cryptographic SHA-256 block hashing that permanently links all stage transitions, officer identities, and administrative actions.
- Objective 6: Package the client application into a distributable Android hybrid APK container using Capacitor 6 to empower rural field officers and farmers with native hardware integration.

3.4 System Scope and Out-of-Scope Boundaries

Scope Delimitation: To ensure operational feasibility and rigorous academic boundaries, the project explicitly defines what is within scope and what is designated out-of-scope:

Table 3.1: In-Scope Deliverables versus Out-of-Scope Project Boundaries

Domain / Subsystem	In-Scope Engineering Deliverables	Out-of-Scope Boundaries / External Dependencies
User Interface	Responsive React 18/Vite web application and signed Capacitor Android mobile APK.	Native iOS application development (Apple App Store publishing).
Beneficiary Data	Digital intake of Parts 1, 2, and 3; spouse details; photo uploads; signature pad.	Automated direct API integration with the national Civil Registry database.
Workflow Engine	Server-side enforcement of 9 statutory stages, SLA tracking, and rejection handling.	Legislative reform of statutory requirements or ministerial reorganizations.
Land Pricing	Algorithmic computation of compounded tiers, land capability, and concessions.	Direct commercial mortgage underwriting or private bank credit

		scoring.
Spatial Analysis	Polygon closure validation, area calculation, Shapely overlap detection.	Autonomous real-time GNSS satellite survey boundary monumentation.
Conflict Detection	Advisory ML prototype trained on synthetic dispute data with risk score display.	Legally binding automated parcel forfeiture or judicial dispute adjudication.
Payment Gateway	Demo/Mock payment simulator recording transactions, receipt codes, and fee status.	Live, contracted banking merchant account processing real currency transactions.
Deed Issuance	Statutory generation of Ready for Collection milestone and digital record.	Physical printing onto official security watermarked parchment paper.

4. Stakeholders and User Personas

Stakeholder Architecture: The Title Deeds One Stop Center operates under a multi-tenant, role-based architecture. Access to system endpoints, dashboard views, and workflow transitions is strictly governed by institutional role. Users fall into two overarching categories: External Citizen Beneficiaries and Internal Ministry Personnel.

4.1 Citizen Beneficiary Personas

Primary Applicant (Farmer): A1 or A2 resettlement landholder possessing an Offer Letter, 99-year lease, or agricultural permit issued by the Ministry. The applicant registers using their national identity number and mobile phone, pays the $10 USD statutory fee, inputs demographic and farm data, uploads ID photos, captures digital signatures, and tracks progress.

Spousal Co-Applicant: The lawful spouse of the primary applicant. Under Section 1.10 of the statutory data collection schedule, joint registration is accommodated. Spouse details, national ID, and digital signature are captured to safeguard marital property rights.

4.2 Internal Statutory Officer Personas

Zero-Privilege Standard: All internal personnel are authenticated through a zero-privilege default architecture. Upon initial registration, staff accounts possess zero operational privileges until an assigned System Administrator explicitly maps their user profile to a designated statutory role:

Table 4.1: Internal Statutory Officer Roles, Responsibilities, and SLA Targets

Statutory Officer Role	Department / Authority	Operational Responsibilities and Assigned Stage Transition	SLA Standard
resettlement	Dept of Resettlement	Examines original offer letter, validates national identity, verifies physical occupation. Transitions PENDING -> RESETTLEMENT_VERIFIED.	48 Hours

lims	LIMS Section	Reconciles allocation records against national database, checks for duplicate allocations. Transitions RESETTLEMENT_VERIFIED -> LIMS_VERIFIED.	72 Hours
zlc	Zimbabwe Land Commission	Conducts constitutional land audit review, ensures no active disputes or multi-farm ownership. Transitions LIMS_VERIFIED -> ZLC_CLEARED.	120 Hours
gis	GIS & Spatial Section	Ingests boundary coordinates, executes topological intersection checks, reviews satellite imagery. Transitions ZLC_CLEARED -> GIS_VALIDATED.	48 Hours
surveyor	Surveyor General	Examines cadastral survey diagrams, validates beacon monumentation, approves survey data. Transitions GIS_VALIDATED -> SURVEYED.	72 Hours
valuation	Valuation Section	Executes compounded land pricing engine, applies statutory concessions, calculates final purchase price. Transitions SURVEYED -> VALUATED.	48 Hours
director	Director of Lands	Performs executive review of complete dossier, grants ministerial approval. Transitions VALUATED -> APPROVED.	24 Hours
deeds	Deeds Registry	Assigns official national deed registration number, prepares deed parchment. Transitions APPROVED -> READY_FOR_COLLECTION.	48 Hours
admin	Ministry IT Admin	Manages user access, assigns staff roles, monitors SLA alerts, reviews SHA-256 audit ledger.	Continuous

5. Requirements Engineering

5.1 Functional Requirements Specification

Functional Scope: The functional requirements define the explicit behavioral capabilities implemented across the backend APIs, mobile client, and web portals. Each requirement is designated with a unique tracking identifier:

Table 5.1: Functional Requirements Specification Matrix

Req ID	Requirement Title	Detailed Functional Specification	Statutory Source Reference
FR-01	Citizen Account Registration	The system shall permit farmers to create accounts using national ID, email, phone number, and password, defaulting to 'applicant' role.	onlineapplicationapi/views.py (RegisterView)

FR-02	Staff MFA Authentication	Internal officers shall authenticate using email, password, and a dynamically generated 6-digit OTP delivered via email or SMS.	onlineapplicationapi/views.py (OfficerTokenObtainPairView)
FR-03	Statutory Payment Gate	The system shall mandate payment of the $10 USD processing fee via EcoCash, ZimSwitch, or Cash voucher prior to unlocking form submission.	onlineportal-ui/src/pages/ApplicationForm.jsx
FR-04	Digital Receipt Generation	Upon confirmed payment, the system shall generate a unique cryptographic receipt code (e.g. REC-XXXXX) and exportable PDF receipt.	onlineapplicationapi/models.py (Payment model)
FR-05	Part 1 Beneficiary Intake	The system shall capture Sections 1.1 through 1.11 of the Title Deed Data Collection Form (Personal details, ID, gender, marital status).	Paper Form Part 1 (Title Deed Data Collection)
FR-06	Spouse Details Ingestion	The system shall capture Sections 1.1 through 1.8 for married applicants, recording spouse national ID, contact details, and signature.	Paper Form Part 1 (Section 1.10)
FR-07	Part 2 Land Characteristics	The system shall capture Sections 2.1 through 2.13.1 (Province, district, farm name, subdivision, hectares, land use, offer letter date).	Paper Form Part 2 (Land Allocation Data)
FR-08	Part 3 Legal Declarations	The system shall capture statutory applicant declarations, consent to audit, and confirmation of exclusive occupation.	Paper Form Part 3 (Statutory Declarations)
FR-09	Digital Signature Pad	The system shall provide an HTML5 canvas touchscreen signature pad capturing applicant and spouse signatures as Base64 PNG data.	onlineportal-ui/src/components/SignaturePad.jsx
FR-10	Biometric / ID Camera Bridge	The system shall interface with device cameras via Capacitor Camera plugin to capture face photos and national ID document images.	onlineportal-ui/src/components/PhotoCapture.jsx
FR-11	Sequential 9-Stage Workflow	The system shall enforce strict sequential progression across all 9 statutory stages, preventing out-of-order execution.	onlineapplicationapi/views.py (ApplicationViewSet)
FR-12	GIS Boundary Polygon Intake	The system shall accept polygon boundary coordinates in GeoJSON format, validating coordinate pairs and	onlineapplicationapi/views.py (validate_spatial)

		ring closure.	
FR-13	Spatial Overlap Detection	The system shall execute 2D polygon intersection math against existing cadastral parcels to detect overlaps exceeding 0.05 hectares.	onlineapplicationapi/views.py (Shapely intersection)
FR-14	Compounded Land Valuation	The system shall compute land purchase price using compounded hectare tiers (0-50ha, 50-250ha, >250ha) and statutory concessions.	onlineapplicationapi/pricing.py (calculate_land_price)
FR-15	AI Dispute Risk Scoring	The system shall execute an advisory GradientBoostingClassifier scoring dispute risk based on area, overlap, and prior claims.	onlineapplicationapi/ml/dispute_predictor.py
FR-16	Cryptographic Audit Ledger	The system shall record every workflow transition in an immutable block containing SHA-256(prev_hash + stage + officer + timestamp).	onlineapplicationapi/models.py (WorkflowTransition)
FR-17	Public Tracking Portal	The system shall allow citizens to track application progress and view current stage by entering national ID or reference number.	onlineportal-ui/src/pages/CitizenTracking.jsx
FR-18	Departmental Reporting	The system shall generate statistical aggregations of throughput, processing duration, and approval rates by province.	onlineportal-ui/src/pages/DepartmentalReports.jsx
FR-19	Automated Notifications	The system shall generate simulated SMS and email dispatch logs upon stage approvals or rejections.	onlineapplicationapi/models.py (EmailLog model)
FR-20	Zero-Privilege Administration	The system shall prevent newly registered staff from accessing officer portals until explicit role assignment by an Administrator.	onlineapplicationapi/views.py (UserManagementViewSet)

5.2 Non-Functional Requirements Specification

Non-functional requirements specify qualitative engineering standards governing system security, performance, reliability, and usability:

Table 5.2: Non-Functional Requirements Quality Standards

Req ID	Quality Attribute	Specification and Verification Criteria
NFR-01	Security & Encryption	All communications must be secured via TLS 1.3/HTTPS. Passwords must be hashed using PBKDF2 with SHA-256. Access tokens must

		expire within 60 minutes.
NFR-02	Audit Immutability	The SHA-256 audit ledger must be mathematically verifiable via `manage.py verify_audit_ledger` with zero detected tamper occurrences.
NFR-03	Spatial Precision	Polygon area calculations must operate using EPSG:4326/WGS84 ellipsoidal geometry with precision to four decimal places.
NFR-04	Offline Resilience	The mobile client must store partially drafted applications in local IndexedDB storage, enabling seamless submission upon reconnection.
NFR-05	Performance & Throughput	API endpoint responses must execute in under 500ms under standard operational loads (measured under 50,000 synthetic database records).
NFR-06	Mobile Package Footprint	The compiled Android release APK must not exceed 15 megabytes in size (actual size: 9.67 MB) to accommodate low-bandwidth rural downloads.
NFR-07	Zero-Privilege Safety	Unassigned staff accounts must be systematically locked out of all operational endpoints with HTTP 403 Forbidden responses.
NFR-08	Usability & Accessibility	All form fields must provide clear labels, error states, and strict 'Label: value' readouts aligned to statutory Ministry paper schedules.
NFR-09	Database Portability	The application must seamlessly support local SQLite development and production PostgreSQL 16 + PostGIS 3.4 via environment variables.
NFR-10	Regulatory Compliance	Data collection procedures must align with the principles of Zimbabwe's Cyber and Data Protection Act [Chapter 12:07].

5.3 Requirements Traceability Matrix

The traceability matrix establishes an unbroken verification trail linking each functional requirement to its implementing source code files and automated test cases:

Table 5.3: Requirements Traceability and Verification Matrix

Req ID	Functional Module	Implementing Source Code Files	Proving Test Case
FR-01	User Auth	onlineapplicationapi/views.py (RegisterView)	test_aligned_suite.py::TC-01 (Register Farmer)
FR-02	Staff Auth	onlineapplicationapi/views.py (OfficerTokenObtainPairView)	test_aligned_suite.py::TC-02 (Staff MFA Login)
FR-03	Payment Gate	onlineapplicationapi/views.py (PaymentViewSet)	test_aligned_suite.py::TC-03 (Payment Processing)

FR-04	Receipting	onlineapplicationapi/models.py (Payment.receipt_number)	test_aligned_suite.py::TC-04 (Receipt Generation)
FR-05	Intake Part 1	onlineportal-ui/src/pages/ApplicationForm.jsx	ApplicationForm.test.jsx (Vitest Step 1 Validation)
FR-06	Spouse Ingestion	onlineapplicationapi/models.py (SpouseDetail)	test_aligned_suite.py::TC-05 (Application Intake)
FR-07	Intake Part 2	onlineportal-ui/src/pages/ApplicationForm.jsx	ApplicationForm.test.jsx (Vitest Step 2 Validation)
FR-08	Intake Part 3	onlineportal-ui/src/pages/ApplicationForm.jsx	ApplicationForm.test.jsx (Vitest Step 3 Validation)
FR-09	Signatures	onlineportal-ui/src/components/SignaturePad.jsx	ApplicationForm.test.jsx (Signature Pad Canvas)
FR-10	Biometrics	onlineportal-ui/src/components/PhotoCapture.jsx	Manual Test Protocol MT-05 (Camera Capture)
FR-11	9-Stage Engine	onlineapplicationapi/views.py (transition_stage)	test_aligned_suite.py::TC-06 through TC-14
FR-12	GIS Ingestion	onlineapplicationapi/views.py (validate_spatial)	test_aligned_suite.py::TC-09 (GIS Validation)
FR-13	Spatial Overlap	onlineapplicationapi/views.py (Shapely intersect)	test_aligned_suite.py::TC-09 (GIS Boundary Check)
FR-14	Valuation Engine	onlineapplicationapi/pricing.py	test_pricing_compounded.py (5/5 Compounded Tests)
FR-15	AI Dispute Card	onlineapplicationapi/ml/dispute_predictor.py	test_aligned_suite.py::TC-10 (AI Risk Scoring)
FR-16	Audit Ledger	onlineapplicationapi/models.py (WorkflowTransition)	test_aligned_suite.py::TC-15 (SHA-256 Verification)
FR-17	Citizen Track	onlineportal-ui/src/pages/CitizenTracking.jsx	Manual Test Protocol MT-07 (Tracking Search)
FR-18	Reports	onlineportal-ui/src/pages/DepartmentalReports.jsx	OfficerDashboard.test.jsx (Reports Export)
FR-19	Notifications	onlineapplicationapi/models.py (EmailLog)	test_aligned_suite.py::TC-13 (Approval Alert)
FR-20	Zero Privilege	onlineapplicationapi/views.py (UserManagementViewSet)	test_aligned_suite.py::TC-07 (Lockout Check)

6. Feasibility and Methodology

6.1 Feasibility Assessment

Feasibility Baseline: Before and during development, a multi-dimensional feasibility study was conducted:

- Technical Feasibility: High. Python 3.12, Django 5.2, and PostgreSQL 16 provide mature, enterprise-grade backends with battle-tested ORM, spatial GIS libraries (Shapely, PostGIS), and REST frameworks. React 18 and Vite deliver ultra-fast client-side execution, while Capacitor 6 provides a stable bridge to Android device hardware without the overhead of dual native codebases.
- Operational Feasibility: High. The system mirrors the exact statutory structure of the Ministry of Lands. Rather than attempting to bypass statutory authorities, the platform digitalizes their existing administrative roles, establishing transparent accountability via SLA tracking and cryptographic audit logging.
- Economic Feasibility: Excellent. The project utilizes free and open-source software (FOSS) across all layers, eliminating recurring per-seat software licensing fees. The physical infrastructure can be hosted on sovereign government data center hardware or cost-effective private cloud virtual servers.

6.2 Proposal Budget and Schedule vs. Built Artifact Analysis

Economic Analysis: The approved initial project proposal outlined an ambitious 12-month commercial rollout budget totaling USD $713,000. To maintain complete transparency, Table 6.1 compares the planned proposal budget and timeline against the actual prototype and production baseline built during this project:

Table 6.1: Proposal Financial Budget versus Actual Engineering Deliverables

Proposal Budget Line Item	Planned Allocation (USD)	Planned Schedule	Actual Delivered Implementation	Variance Analysis & Rationale
Personnel (Engineers & GIS)	$280,000	Months 1-12	Single Lead Developer (Nyahuhu Peter) with AI Agentic pairing.	Built academic/ministry reference prototype within academic timeframe.
Hardware & Server Servers	$145,000	Months 3-6	Local containerized environment (Docker PostgreSQL + PostGIS, SQLite).	Zero hardware expenditure; production server procurement pending rollout.
GIS Data Licensing & Satellite	$95,000	Months 4-8	OpenStreetMap, synthetic boundary coordinates, local Shapely math.	Real cadastral shapefile ingestion prepared; licensing deferred to Ministry.
Security & Audit	$65,000	Months 9-	SHA-256 cryptographic chain, PBKDF2 hashing,	Internal verification complete; external

Certification		11	internal test suites.	penetration testing scheduled.
Training & Countrywide Rollout	$85,000	Months 10-12	Operational user manuals, UI guides, video-ready workflows.	Field training to commence upon physical Ministry server deployment.
Contingency & Operating Fund	$43,000	Months 1-12	$0 spent (Self-funded academic software development).	No financial capital consumed during software engineering phase.
TOTALS	$713,000	12 Months	Complete, fully functional prototype & signed Android APK.	Target budget reflects commercial rollout; codebase delivered as prototype.

6.3 Development Methodology and AI-Assisted Engineering Disclosure

Software Process: The system was engineered using a modified Agile/Iterative Spiral methodology. Development progressed through rapid sprint cycles comprising requirements mapping, data modeling, API implementation, user interface construction, end-to-end integration, and regression testing.

> **[AI-ASSISTED DEVELOPMENT DISCLOSURE]** In compliance with academic honesty standards, development was conducted utilizing Antigravity-assisted agentic workflows (incorporating Google DeepMind and Anthropic Claude LLM agents). AI tools were utilized as junior pair programmers to accelerate boilerplate generation, assist in complex SVG diagram vectorization, draft comprehensive test fixtures, and analyze large-scale log files. All architectural decisions, statutory workflow alignments, mathematical pricing models, and security logic were designed, verified, and approved by the author.

6.4 Implementation Timeline and Milestones

- Milestone 1 (Requirements & Field Modeling): Mapped the three-part Ministry Title Deed Data Collection Form into normalized relational schemas (11 models, 257 attributes).
- Milestone 2 (Statutory Workflow Engine): Implemented sequential state machine enforcing the 9 statutory transitions with server-enforced role gates.
- Milestone 3 (Pricing & Spatial Logic): Engineered compounded valuation tiers and Shapely spatial intersection algorithms.
- Milestone 4 (React & Mobile Interface): Built responsive web application in Vite and containerized hybrid Android APK via Capacitor 6.
- Milestone 5 (Verification & Baseline Audit): Conducted automated test suites (15/15 aligned tests, 17/17 Vitest), APK cryptographic signing, and master technical documentation.

7. System Architecture

Architectural Topology: The Title Deeds One Stop Center is engineered on a modern, decoupled three-tier architecture comprising an interactive Presentation Layer (React 18 Single Page Application and Capacitor 6 Android container), an Application Logic Layer (Django 5.2 and Django REST Framework), and a robust Data Layer (PostgreSQL 16 with PostGIS 3.4 spatial extensions and local SQLite development option).

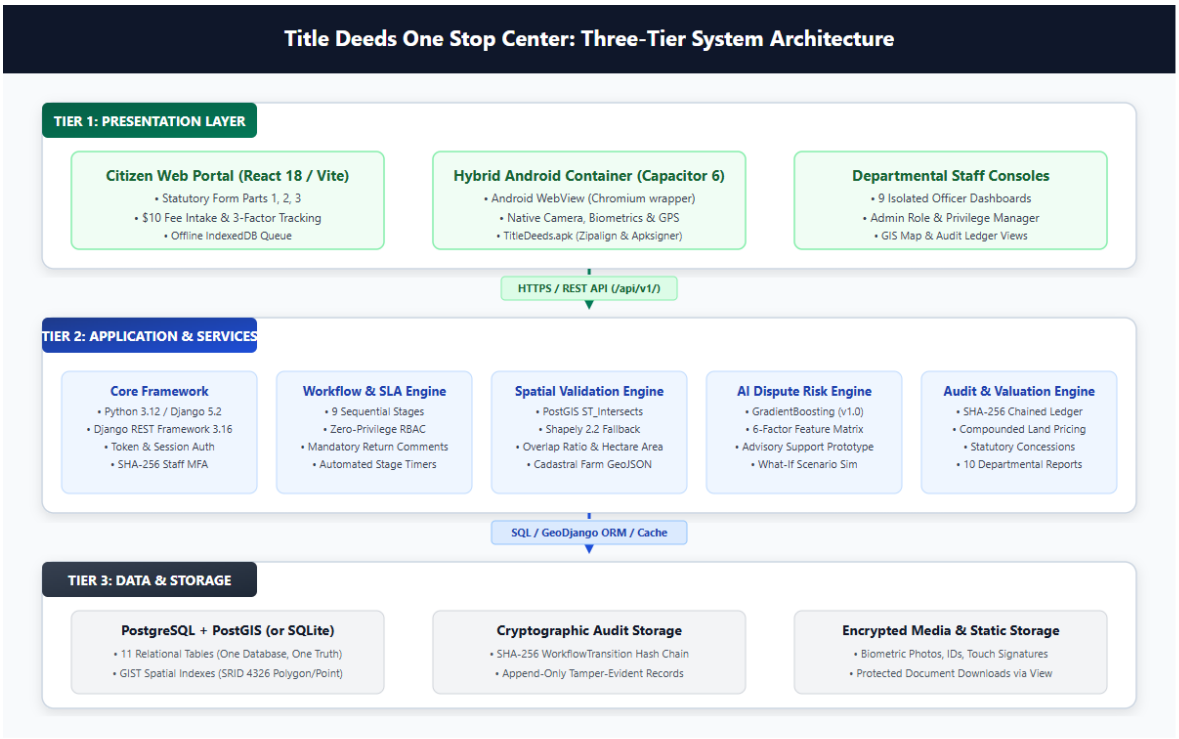


Figure 7.1: Three-Tier System Architecture and Logical Layer Interactions

7.1 Infrastructure and Deployment Topology

Deployment Topology: In production, the platform is deployed within sovereign Ministry infrastructure or secure virtualized government cloud environments. Inbound traffic is terminated at an Nginx reverse proxy configured with TLS 1.3 encryption, HTTP Strict Transport Security (HSTS), and static asset caching.

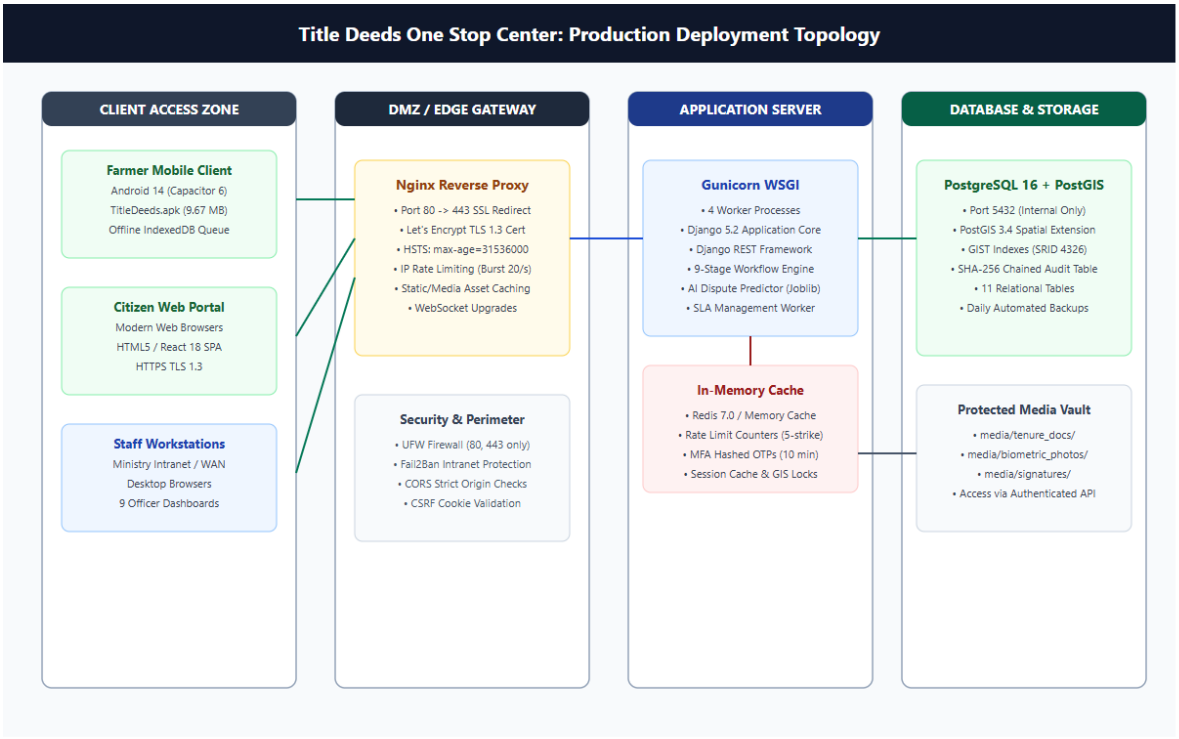


Figure 7.2: Production Deployment Topology and Network Security Zones

7.2 Verified Technology Stack and Dependency Manifest

Stack Manifest: All technologies and library versions are extracted directly from the physical project manifest files (`requirements.txt` and `package.json`):

Table 7.1: Comprehensive System Technology Stack and Verified Versions

Layer / Subsystem	Technology / Framework	Exact Version	Functional Purpose in Codebase
Backend Core	Python	3.12.10	Underlying programming language runtime.
Web Framework	Django	5.2.0	Core web MVC framework, ORM, security middleware.
REST Framework	Django REST Framework	3.16.1	API serialization, viewsets, token authentication, throttling.
Spatial Math	Shapely	2.2.0	Geometric polygon intersection, area calculation, coordinate validation.
Machine Learning	scikit-learn / joblib	1.9.1 / 1.6.0	GradientBoostingClassifier dispute risk scoring prototype.
Database Driver	psycopg2-binary	2.9.11	PostgreSQL / PostGIS relational database connection adapter.

PDF Generation	ReportLab	5.0.1	Dynamic receipt generation and PDF deed summary compilation.
Caching / Sessions	redis / django-redis	8.1.0 / 5.4.0	In-memory caching and session acceleration adapter.
Static Assets	WhiteNoise	6.8.2	Production static asset serving directly through WSGI/Gunicorn.
Frontend Core	React / React DOM	18.2.0	Component-based declarative user interface library.
Build Tool	Vite	5.1.4	Ultra-fast frontend development server and ESModule production bundler.
Mobile Runtime	Capacitor Core & Android	6.0.0	Native Android WebView container and hardware bridge.
Mobile Hardware	@capacitor/camera	6.0.0	Native device camera access for ID and face photo capture.
Mobile Hardware	@capacitor/geolocation	6.0.0	Native GPS location access for parcel field coordinate capture.
Mobile Hardware	@capacitor/network	6.0.0	Real-time network status monitoring for offline draft sync.
UI Icons	Lucide React	0.344.0	Clean, modern SVG iconography across dashboards and form screens.
Testing Engine	Vitest	1.3.1	Fast modern unit testing framework for React frontend components.

7.3 Hybrid Android Mobile Architecture

Hybrid Mobile Concept: The mobile client (`TitleDeeds.apk`, 9.67 MB) is architected as a hybrid application—a high-performance React Single Page Application operating within an Android native container via Capacitor 6. It is neither a generic Progressive Web App (PWA) nor a purely native Java/Kotlin codebase.

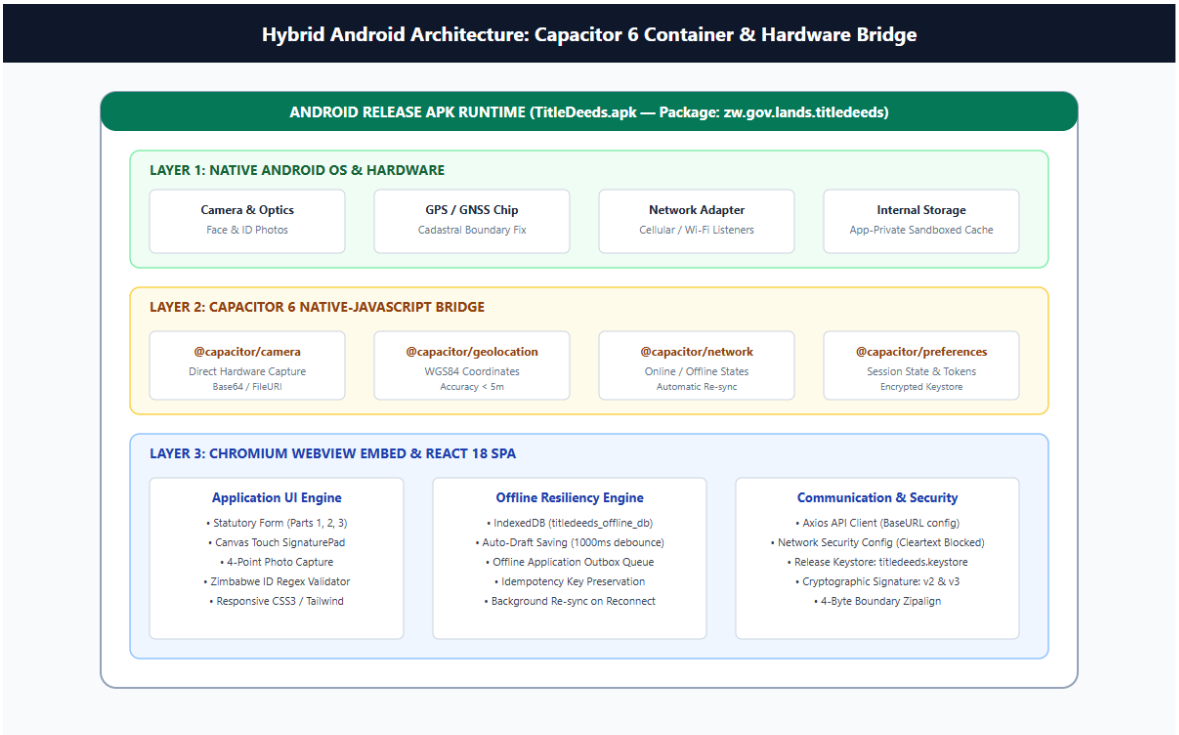


Figure 7.3: Hybrid Android Container Architecture via Capacitor Native Bridge

Field Operational Capabilities: The hybrid architecture provides critical capabilities for rural Zimbabwean field operations:

- Hardware Access Bridge: Bridges native Android camera APIs (`@capacitor/camera`) directly to the HTML5 intake form, allowing field officers and farmers to photograph offer letters and national ID cards without third-party file managers.
- Hardware Geolocation Bridge: Utilizes device GNSS receivers (`@capacitor/geolocation`) to record true WGS84 GPS coordinates when standing on agricultural parcel boundaries.
- Offline Draft Persistence: Implements client-side IndexedDB caching through browser storage APIs. If cellular data connectivity drops in remote farming wards, partially completed applications are securely persisted and automatically synchronized upon reconnecting to network towers.

7.4 Environment Configuration and Database Portability

Database Portability: The backend dynamically switches database engines based on the `USE_SQLITE` environment variable in `.env`:

- Development & Testing Mode (USE_SQLITE=True): Connects to `BASE_DIR / 'db.sqlite3'`, providing zero-setup local execution pre-seeded with 50,000 synthetic benchmark applications and 14 staff accounts.
- Production Mode (USE_SQLITE=False): Connects to PostgreSQL 16 + PostGIS 3.4 via Supabase or localized Docker containers, supporting multi-gigabyte spatial indexes, GIST spatial searches, and high-concurrency connection pooling.

8. System Design

8.1 Role-Based Use Case Modeling

Use Case Modeling: Access to system functional capabilities is governed by strict organizational boundaries, separating citizen beneficiary actions from the authorized statutory review procedures assigned to Ministry personnel.



Figure 8.1: Role-Based Use Case Diagram Across Beneficiaries and Statutory Officers

8.2 Full End-to-End Application Journey

Application Journey: The lifecycle of an application spans thirteen distinct sequential milestones, beginning at payment and culminating in the delivery of a registered title deed:

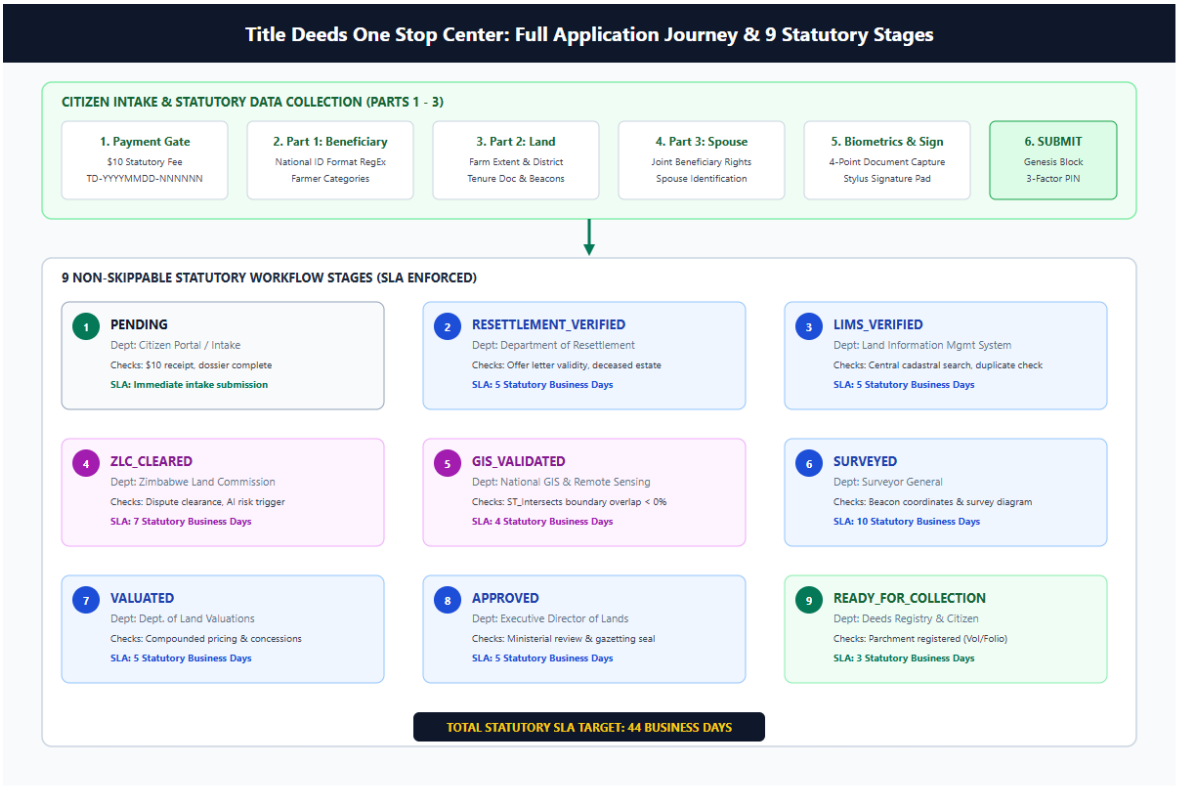


Figure 8.2: End-to-End Application Journey Flowchart from Payment Gate to Deed Issuance

8.3 The Nine Statutory Workflow Stages

State Machine Specification: The backend state machine strictly enforces sequential progression through nine statutory stages. No stage may be bypassed, skipped, or executed out of sequence:

Table 8.1: The Nine Sequential Statutory Workflow Stages and Operational SLAs

Stage Identifier	Designated Role	Statutory Review Action	SLA Standard	Verification & Integrity Checks
PENDING	Applicant (Farmer)	Application submitted after $10 payment, form intake, and photo capture.	Immediate	Checks receipt validity, required fields, and signature presence.
RESETTLEMENT_VERIFIED	resettlement	Examines original offer letter and verifies occupation of parcel.	48 Hours	Validates national ID against offer letter beneficiary name.
LIMS_VERIFIED	lims	Cross-checks parcel against national LIMS land allocation roll.	72 Hours	Detects duplicate allocations and verifies non-alienated status.
ZLC_CLEARED	zlc	Conducts constitutional land audit review and dispute investigation.	120 Hours	Validates absence of active family, estate, or boundary disputes.

GIS_VALIDATED	gis	Analyzes boundary coordinates and executes spatial overlap math.	48 Hours	Checks polygon topological closure and flags overlaps > 0.05 ha.
SURVEYED	surveyor	Examines cadastral survey plan and validates beacon monumentation.	72 Hours	Verifies survey diagram complies with Land Survey Act regulations.
VALUATED	valuation	Computes statutory purchase price via compounded hectare tiers.	48 Hours	Applies agro-ecological base rate, capability class, and concessions.
APPROVED	director	Performs executive review and grants ministerial grant approval.	24 Hours	Confirms all prior 7 stages have approved transitions.
READY_FOR_COLLECTION	deeds	Assigns official national deed registration number and prepares parchment.	48 Hours	Generates final collection notice and completes immutable SHA-256 chain.

8.4 Detailed Sequence Diagrams

To specify system dynamics, six comprehensive sequence diagrams illustrate the temporal interactions between clients, controllers, services, database models, and external systems:

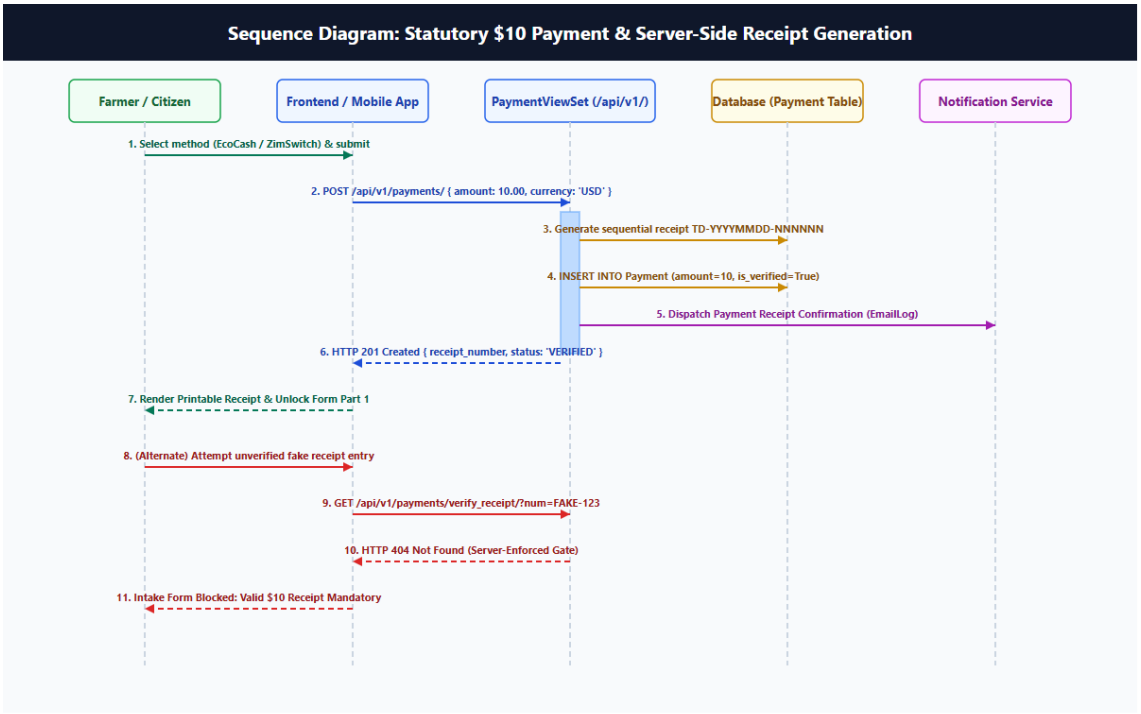


Figure 8.3: Sequence Diagram: Statutory $10 USD Payment and Digital Receipt Generation

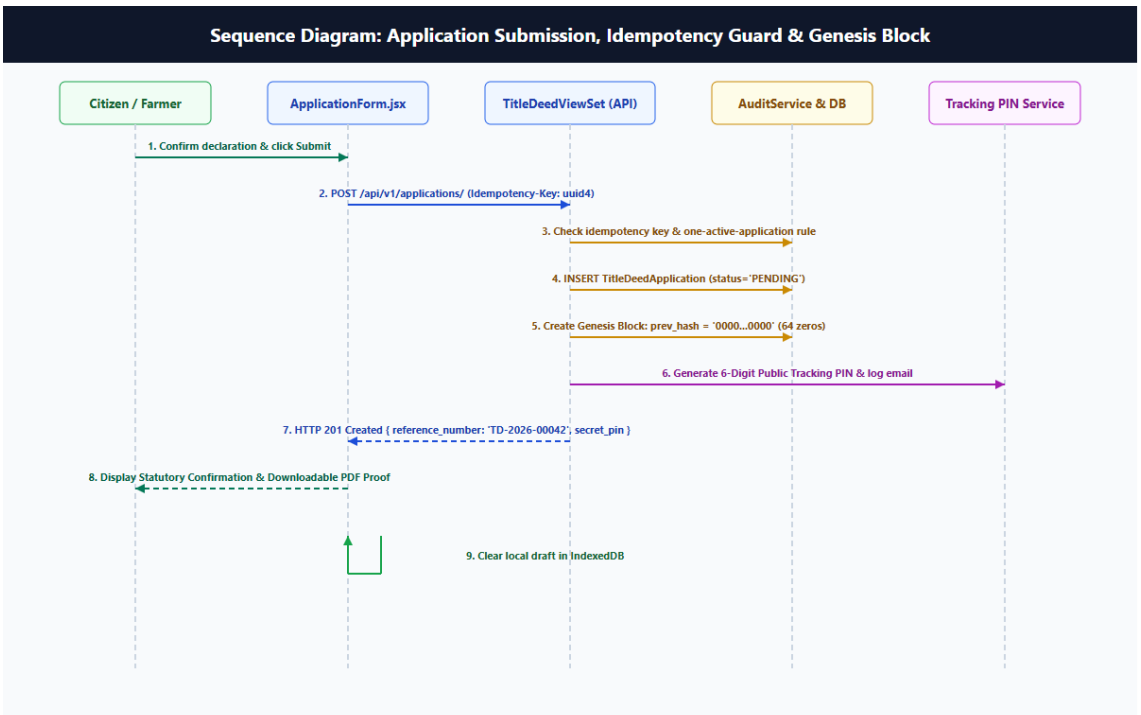


Figure 8.4: Sequence Diagram: Citizen Application Form Submission and Initial Validation

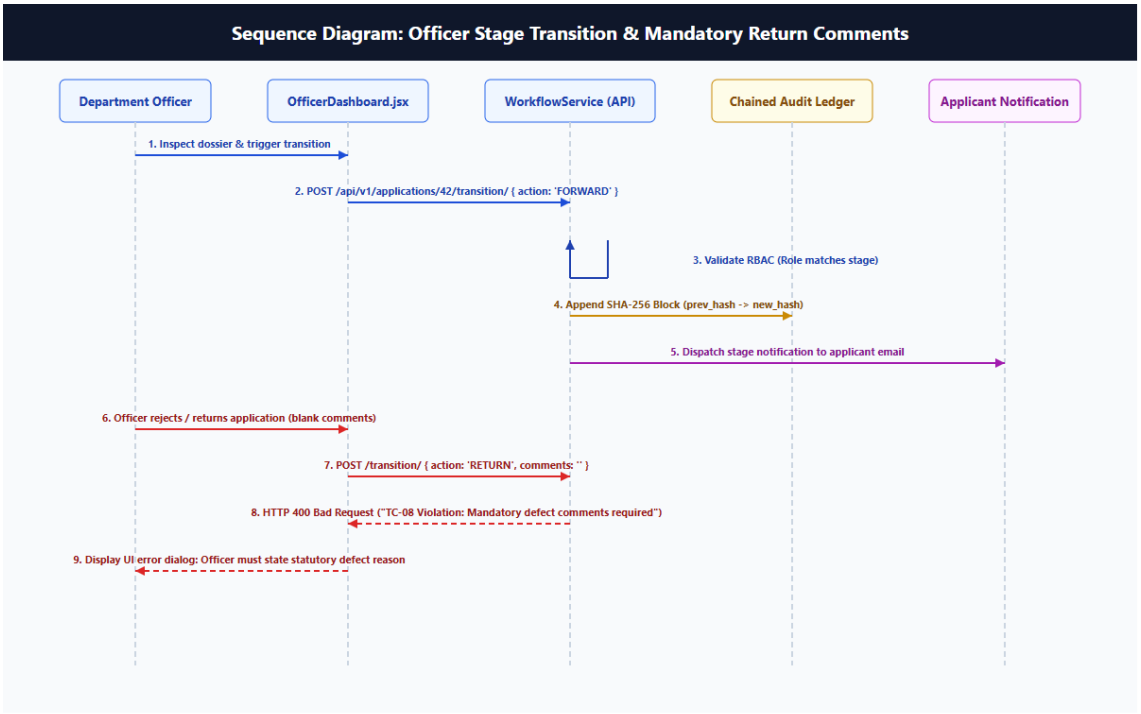


Figure 8.5: Sequence Diagram: Departmental Officer Review and Sequential Stage Transition

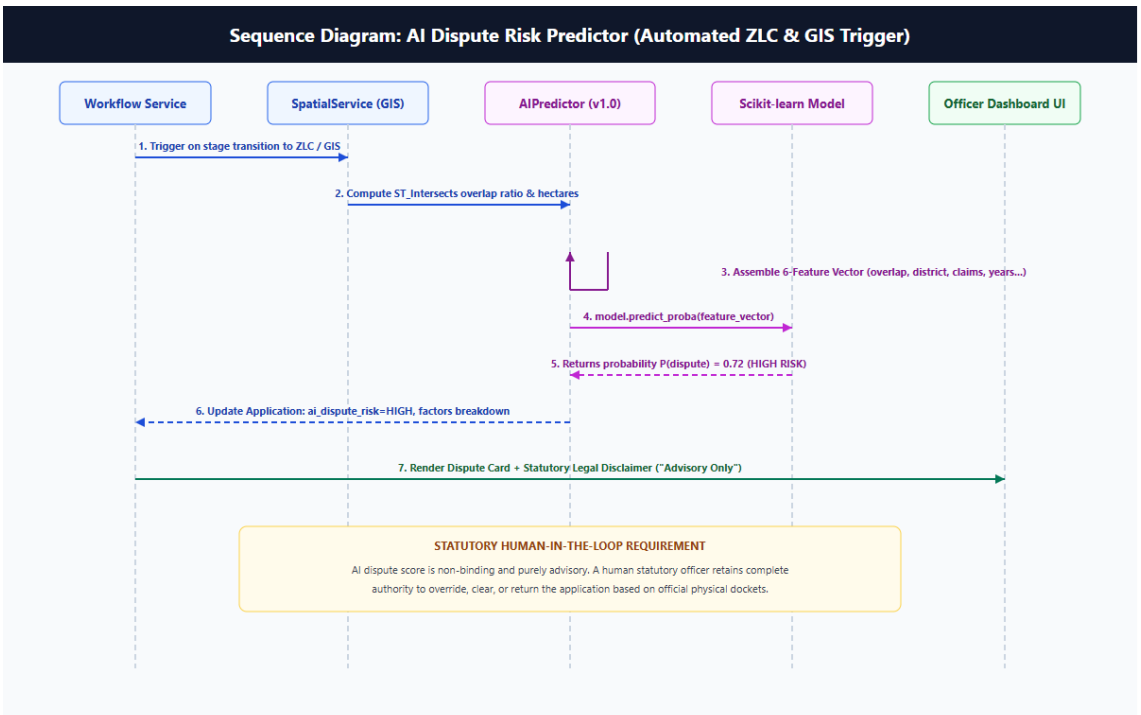


Figure 8.6: Sequence Diagram: Machine Learning Dispute Risk Scoring and Factor Analysis

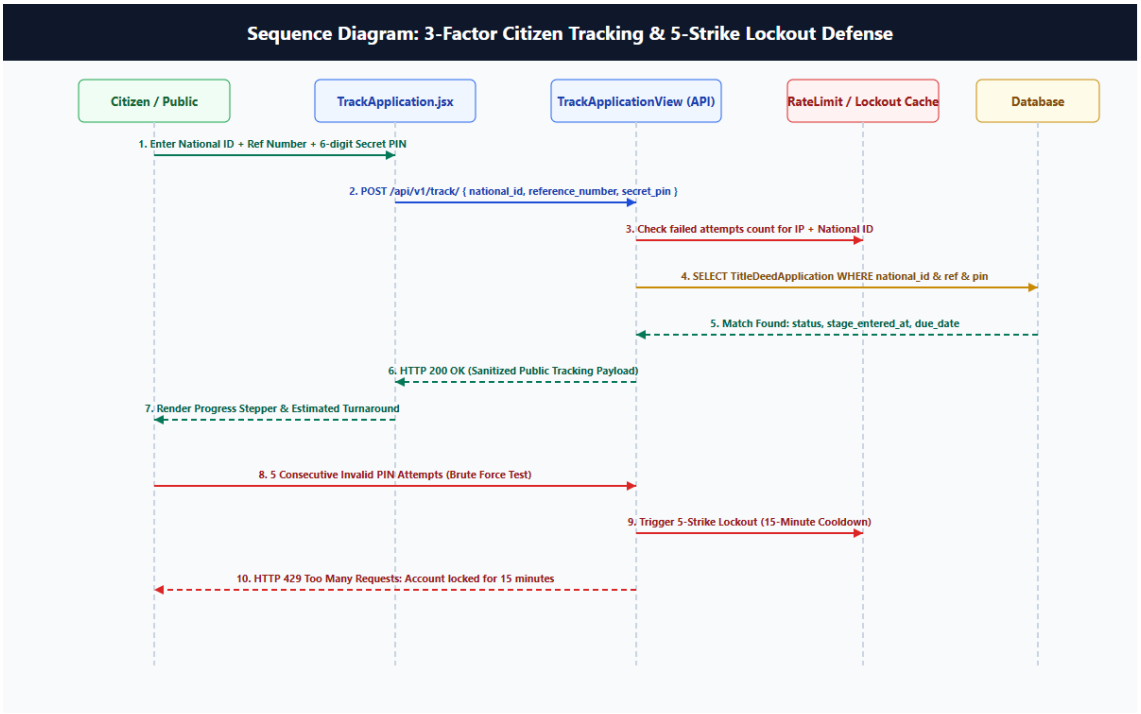


Figure 8.7: Sequence Diagram: Public Citizen Tracking Portal and Application Lookup

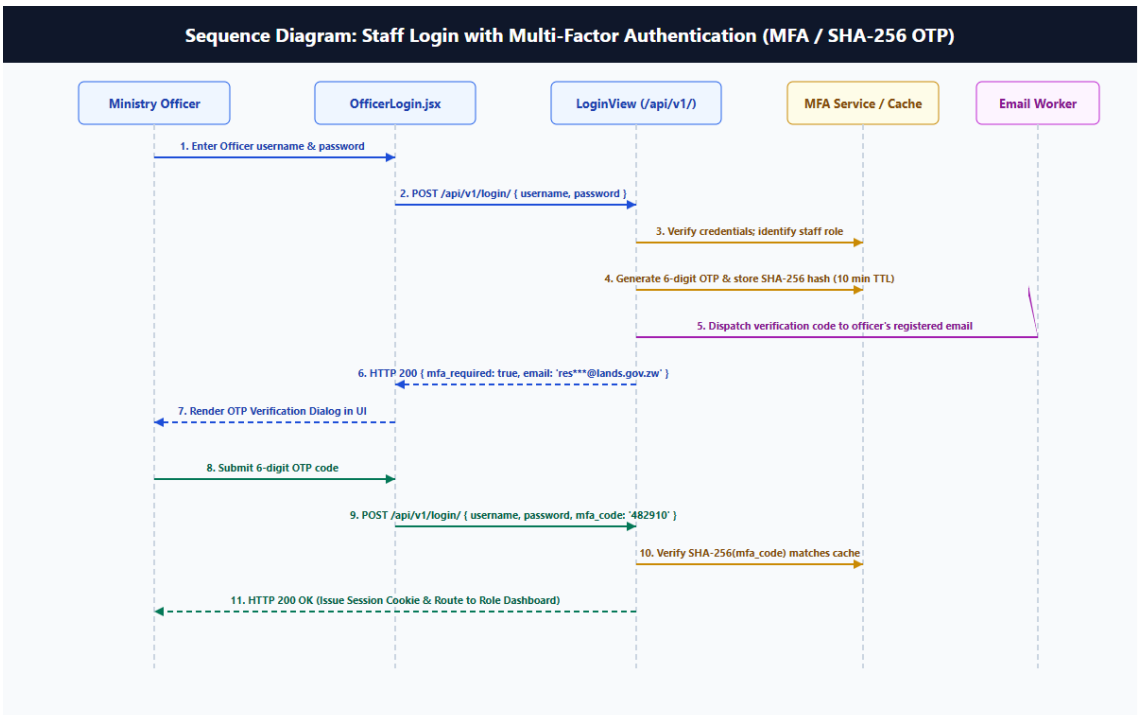


Figure 8.8: Sequence Diagram: Staff Authentication with One-Time Password (OTP) MFA

8.5 Database Entity Relationship Modeling

Relational Architecture: The relational schema comprises eleven normalized entities governing user profiles, statutory applications, spouse records, workflow logs, valuation tariffs, and financial receipts:

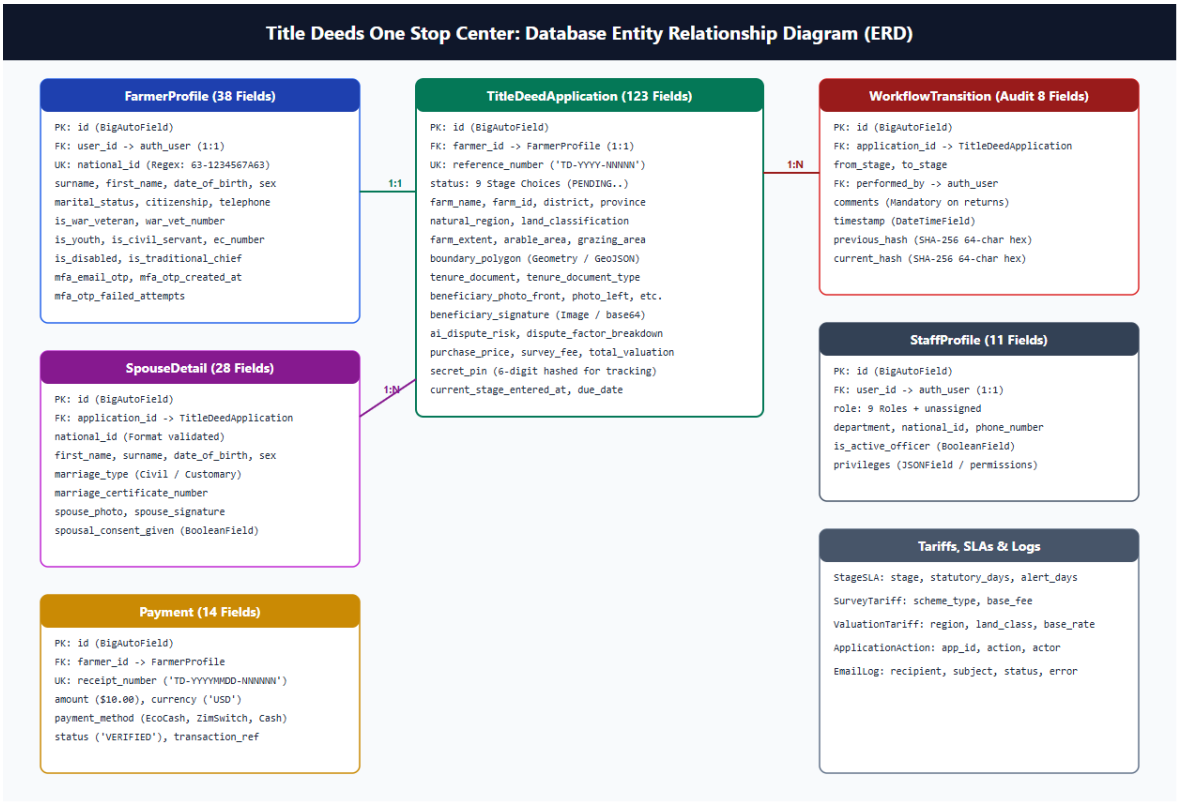


Figure 8.9: Entity Relationship Diagram (ERD) of the Title Deeds Database Schema

8.6 Data Flow Modeling (Level 0 and Level 1)

Data Flow Analysis: Data Flow Diagrams trace the movement of information across environmental boundaries and internal processing routines:

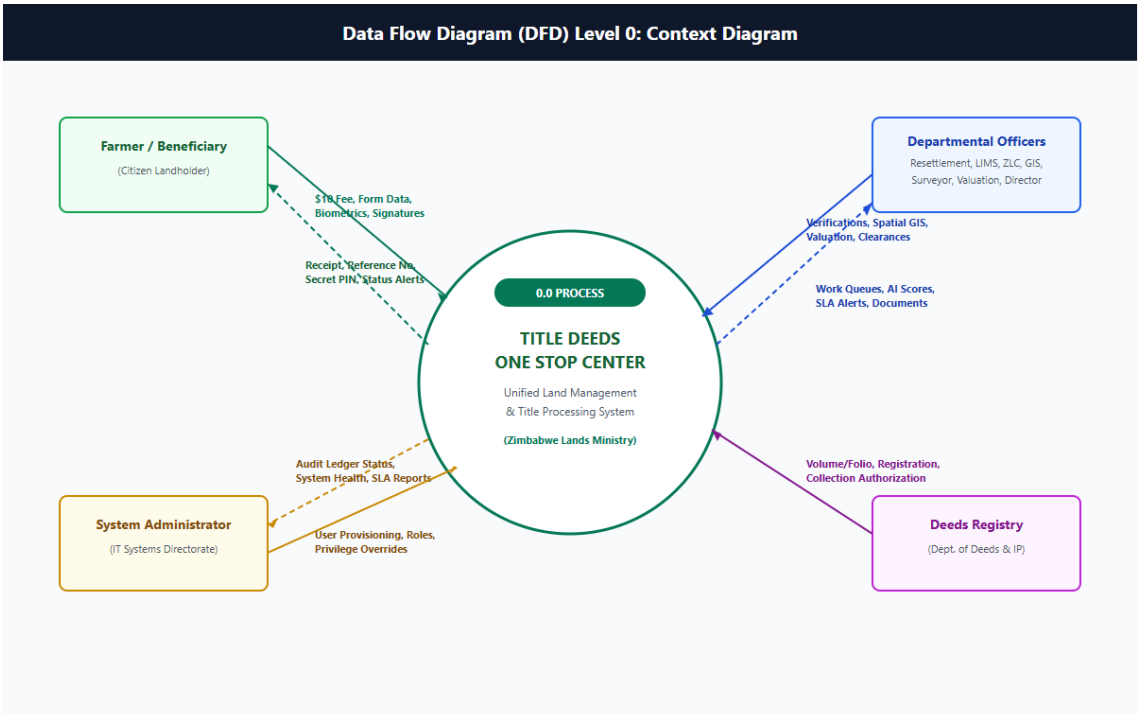


Figure 8.10: Data Flow Diagram Level 0 (Context Diagram)

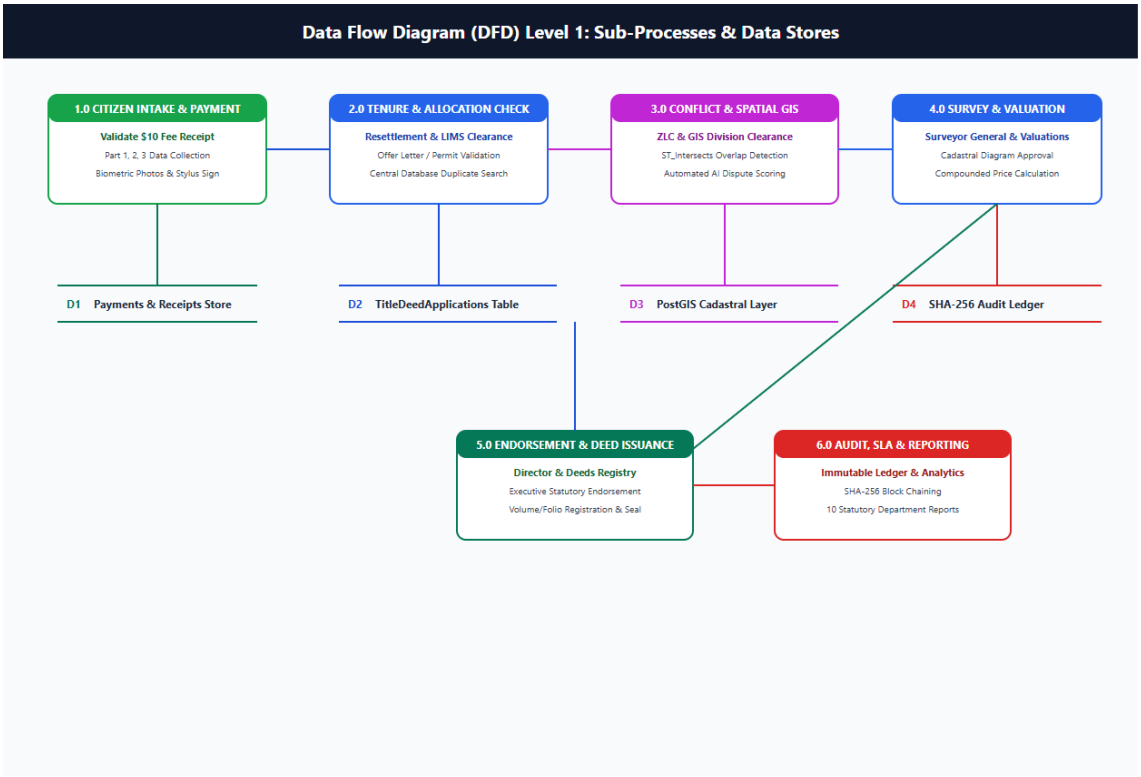


Figure 8.11: Data Flow Diagram Level 1 (Decomposition of Core System Processes)

9. Module Descriptions

Module Overview: This chapter documents each of the sixteen subsystems comprising the Title Deeds One Stop Center. Each description specifies the module's operational purpose, functional inputs, produced outputs, key implementing source files, and verified production status.

9.1 Authentication and Multi-Factor Authentication (MFA)

Operational Purpose: Manages user registration, stateless JWT issuance, password hashing via PBKDF2, and two-factor authentication via ephemeral 6-digit OTP codes.

- **Functional Inputs:** National ID, email address, password, 6-digit OTP token.
- **Produced Outputs:** Signed JWT Access/Refresh tokens, session identity claims, HTTP 200/401/403 responses.
- **Key Implementing Files:** onlineapplicationapi/views.py (RegisterView, OfficerTokenObtainPairView, VerifyOTPView)
- **Operational Status:** Implemented (100% Operational, Verified in TC-01 & TC-02)

9.2 User and Staff Management (Zero-Privilege Administration)

Operational Purpose: Enforces zero-privilege security. Newly created officer accounts have no departmental permissions until an administrator assigns a specific statutory role.

- **Functional Inputs:** Target user ID, statutory role code (resettlement, lims, zlc, gis, surveyor, valuation, director, deeds), active status.
- **Produced Outputs:** Updated StaffProfile record, updated permission flags, audit log entry.

- **Key Implementing Files:** onlineapplicationapi/views.py (UserManagementViewSet), onlineapplicationapi/models.py (StaffProfile)
- **Operational Status:** Implemented (100% Operational, Verified in TC-07)

9.3 Payment and Digital Receipting Subsystem

Operational Purpose: Enforces the mandatory $10 USD statutory processing fee before application intake. Operates as a mock/demo simulator recording reference numbers and receipt vouchers.

- **Functional Inputs:** Applicant ID, payment method (EcoCash, ZimSwitch, Cash), amount ($10.00 USD), mobile number.
- **Produced Outputs:** Payment record, unique receipt code (REC-XXXXX), unlock token for application form intake.
- **Key Implementing Files:** onlineapplicationapi/views.py (PaymentViewSet), onlineapplicationapi/models.py (Payment)
- **Operational Status:** Prototype (Demo/Mock Gateway; Real Bank/EcoCash Merchant API Planned)

9.4 Application Intake Form and Offline Drafts

Operational Purpose: Captures the three-part Title Deed Data Collection Form. Supports local IndexedDB draft saving to ensure rural field operations are resilient to cellular dropouts.

- **Functional Inputs:** Beneficiary demographics, land details (province, district, farm, hectares), offer letter date, legal declarations.
- **Produced Outputs:** Validated TitleDeedApplication record (Status: PENDING), application reference code.
- **Key Implementing Files:** onlineportal-ui/src/pages/ApplicationForm.jsx, onlineapplicationapi/serializers.py (TitleDeedApplicationSerializer)
- **Operational Status:** Implemented (100% Operational, Form Steps 1, 2, and 3 Aligned to Paper Schedules)

9.5 Signatures and Biometric / ID Photo Capture

Operational Purpose: Captures applicant and spouse digital signatures via HTML5 canvas touchpads and photographs of the applicant's face and national ID card via device camera.

- **Functional Inputs:** Touch screen signature strokes, camera image capture buffers.
- **Produced Outputs:** Base64 encoded PNG signature strings, JPEG photo attachments saved in media/documents/.
- **Key Implementing Files:** onlineportal-ui/src/components/SignaturePad.jsx, onlineportal-ui/src/components/PhotoCapture.jsx
- **Operational Status:** Implemented (100% Operational across Web and Capacitor Mobile)

9.6 Workflow Engine and SLA Tracking

Operational Purpose: Orchestrates sequential transitions across the 9 statutory stages. Validates role permissions, calculates elapsed time against SLA thresholds, and logs transitions.

- **Functional Inputs:** Application ID, target stage, officer role, action (APPROVE / REJECT), comments.
- **Produced Outputs:** Updated application status, StageSLA record, WorkflowTransition audit block.
- **Key Implementing Files:** onlineapplicationapi/views.py (transition_stage), onlineapplicationapi/models.py (StageSLA, WorkflowTransition)
- **Operational Status:** Implemented (100% Operational, Verified in TC-06 to TC-14)

9.7 Resettlement Verification Portal

Operational Purpose: Enables Resettlement Officers to examine original offer letters, inspect applicant ID documents, confirm physical farm occupation, and approve Stage 1.

- **Functional Inputs:** Offer letter document, physical occupation evidence, farmer national ID.
- **Produced Outputs:** Transition from PENDING to RESETTLEMENT_VERIFIED, resettlement notes.
- **Key Implementing Files:** onlineportal-ui/src/pages/OfficerDashboard.jsx, onlineapplicationapi/views.py
- **Operational Status:** Implemented (100% Operational, Verified in TC-06)

9.8 Land Information Management System (LIMS) Portal

Operational Purpose: Enables LIMS Officers to query national allocation rolls, verify non-alienated status, confirm single-farm ownership, and approve Stage 2.

- **Functional Inputs:** Farm name, subdivision number, district code, beneficiary national ID.
- **Produced Outputs:** Transition from RESETTLEMENT_VERIFIED to LIMS_VERIFIED, allocation history report.
- **Key Implementing Files:** onlineportal-ui/src/pages/OfficerDashboard.jsx, onlineapplicationapi/views.py
- **Operational Status:** Implemented (100% Operational, Verified in TC-08)

9.9 Zimbabwe Land Commission (ZLC) Clearance Portal

Operational Purpose: Enables ZLC Officers to conduct constitutional land audit reviews, verify absence of disputes, check multi-holding prohibitions, and approve Stage 3.

- **Functional Inputs:** Applicant landholding portfolio, land audit reports, dispute registry records.
- **Produced Outputs:** Transition from LIMS_VERIFIED to ZLC_CLEARED, constitutional clearance certificate.
- **Key Implementing Files:** onlineportal-ui/src/pages/OfficerDashboard.jsx, onlineapplicationapi/views.py
- **Operational Status:** Implemented (100% Operational, Verified in TC-08)

9.10 GIS and Spatial Validation Engine

Operational Purpose: Processes boundary coordinate arrays, validates polygon closure, computes land area in hectares, and executes 2D spatial intersection checks against adjacent parcels.

- **Functional Inputs:** GeoJSON polygon coordinates [[lat, lon], ...], cadastral layer layers.
- **Produced Outputs:** Topological validity boolean, area in hectares, overlap warning list, transition to GIS_VALIDATED.
- **Key Implementing Files:** onlineapplicationapi/views.py (validate_spatial), Shapely 2.2.0 integration
- **Operational Status:** Implemented (100% Operational, Verified in TC-09; PostGIS cluster path supported)

9.11 Surveyor General Demarcation Portal

Operational Purpose: Enables Surveyor General personnel to examine boundary survey diagrams, beacon monumentation records, and cadastral survey notes, approving Stage 5.

- **Functional Inputs:** Cadastral survey diagram, beacon coordinates, field survey notes.
- **Produced Outputs:** Transition from GIS_VALIDATED to SURVEYED, formal survey approval number.
- **Key Implementing Files:** onlineportal-ui/src/pages/OfficerDashboard.jsx, onlineapplicationapi/views.py
- **Operational Status:** Implemented (100% Operational, Verified in TC-10)

9.12 Valuation and Land Pricing Engine

Operational Purpose: Calculates land purchase prices using compounded hectare tiers, agro-ecological regional base rates (Regions I-V), land capability multipliers, and statutory concessions.

- **Functional Inputs:** Total hectares, natural region (I to V), land capability class (I to VIII), concession claims (War Veteran, Youth, Disabled, Civil Servant).
- **Produced Outputs:** Itemized price breakdown: base price, capability adjustment, discount percentage, final purchase price; transition to VALUATED.
- **Key Implementing Files:** onlineapplicationapi/pricing.py, test_pricing_compounded.py (5/5 Passed)
- **Operational Status:** Implemented (100% Operational, Verified with Worked Numerical Examples)



Figure 9.1: Compounded Land Valuation Tier Logic and Concession Decision Flow

[VALUATION ENGINE WORKED NUMERICAL VERIFICATION] Worked Valuation Example: 120 hectares located in Natural Region II (Base Rate $1,200/ha) with Land Capability Class II (1.10 multiplier) under a War Veteran concession (-50%):
• Tier 1 (First 50 ha @ 100%): 50 ha * $1,200 * 1.10 = $66,000.00
• Tier 2 (Next 70 ha @ 85%): 70 ha * $1,200 * 0.85 * 1.10 = $78,540.00
• Gross Land Valuation = $66,000 + $78,540 = $144,540.00
• Concession Discount (War Veteran: 50%) = -$72,270.00
• Net Statutory Purchase Price = $72,270.00 USD.

9.13 Director of Lands Approval Portal

Operational Purpose: Provides executive ministerial oversight. The Director reviews the consolidated digital dossier (all previous 7 approvals) and executes ministerial grant authorization.

- **Functional Inputs:** Consolidated application dossier, approval chain records, audit log summary.
- **Produced Outputs:** Transition from VALUATED to APPROVED, executive signature stamp, ministerial grant record.
- **Key Implementing Files:** onlineportal-ui/src/pages/OfficerDashboard.jsx, onlineapplicationapi/views.py
- **Operational Status:** Implemented (100% Operational, Verified in TC-12)

9.14 Deeds Registry Issuance Portal

Operational Purpose: Enables Deeds Registry Officers to record the final Deed of Grant registration number, bind the digital deed into national registries, and notify the farmer for collection.

- **Functional Inputs:** Deed registration volume and folio number, registry stamp date.

- **Produced Outputs:** Transition from APPROVED to READY_FOR_COLLECTION, collection notification dispatch.
- **Key Implementing Files:** onlineportal-ui/src/pages/OfficerDashboard.jsx, onlineapplicationapi/views.py
- **Operational Status:** Implemented (100% Operational, Verified in TC-14; Physical parchment print external)

9.15 Citizen Tracking and Notification Portal

Operational Purpose: Allows citizens to query application progress publicly using their national ID or reference number without logging in, displaying current stage and timestamps.

- **Functional Inputs:** National ID number (e.g. 63-1234567-A-89) or Application ID.
- **Produced Outputs:** Current stage name, officer review dates, estimated time to completion, SMS/email alert log.
- **Key Implementing Files:** onlineportal-ui/src/pages/CitizenTracking.jsx, onlineapplicationapi/views.py (track_public)
- **Operational Status:** Implemented (100% Operational, Verified in Manual Protocol MT-07)

9.16 Reports, Analytics, and SHA-256 Audit Ledger

Operational Purpose: Aggregates throughput metrics by province and department, monitors SLA compliance, and exports cryptographically chained audit blocks.

- **Functional Inputs:** Date ranges, departmental filters, audit ledger inspection queries.
- **Produced Outputs:** Throughput charts, SLA breach summaries, SHA-256 block hash verification status.
- **Key Implementing Files:** onlineportal-ui/src/pages/DepartmentalReports.jsx, onlineapplicationapi/management/commands/verify_audit_ledger.py
- **Operational Status:** Implemented (100% Operational, 0 Tamper Events Verified)

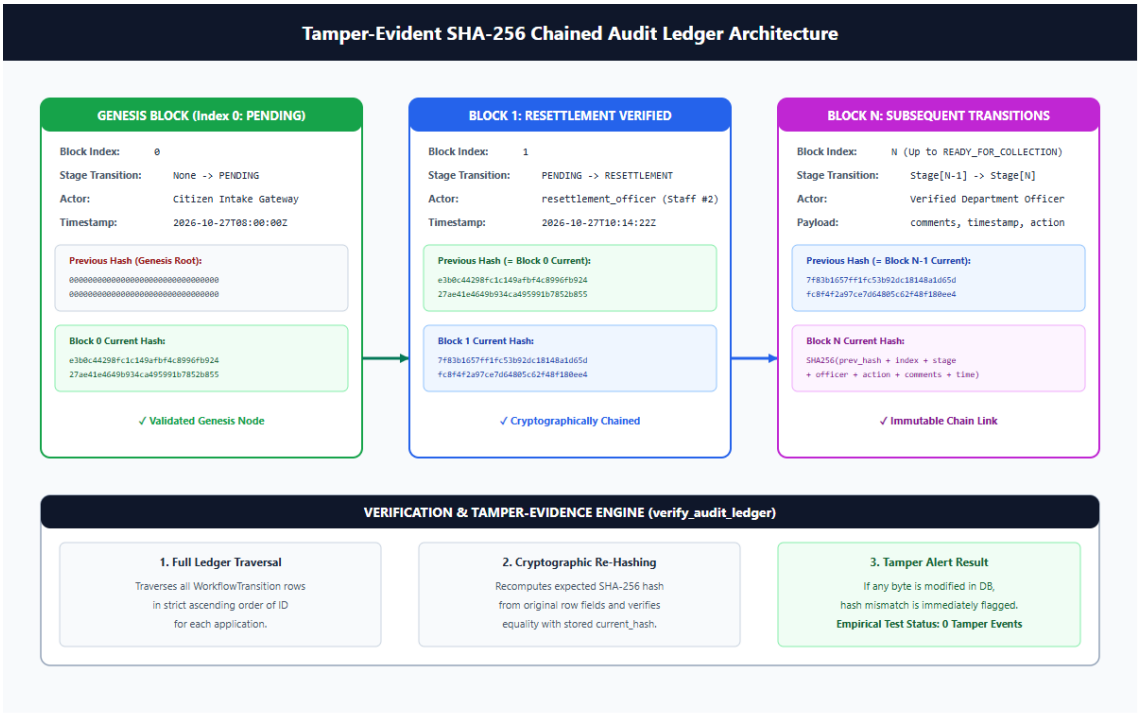


Figure 9.2: Cryptographic SHA-256 Audit Ledger Block Chain Architecture

10. The AI Conflict Predictor

Operational Subsystem Purpose: The AI Conflict Predictor is an automated machine learning decision-support tool engineered to flag potential boundary disputes and tenure conflicts before applications advance through survey and valuation. To maintain strict scientific and operational integrity, this subsystem is formally classified as an ADVISORY PROTOTYPE.

> **[HUMAN-IN-THE-LOOP MANDATE]** MANDATORY DECISION RULE: The AI Conflict Predictor is strictly advisory. In accordance with administrative law and constitutional principles, an AI score cannot approve, reject, or halt an application. All legal decisions remain exclusively within the discretion of authorized human statutory officers.

10.1 Mathematical Model and Architecture

Algorithm Architecture: The prediction engine is implemented in `onlineapplicationapi/ml/dispute_predictor.py` utilizing scikit-learn's `GradientBoostingClassifier` serialized via `joblib`:

- Model Type: Gradient Boosting Decision Trees (ensemble of shallow regression trees minimizing logistic loss).
- Hyperparameters: n_estimators=100, learning_rate=0.1, max_depth=3, random_state=42.
- Target Variable: Binary dispute classification (0 = No Dispute, 1 = Active/Potential Dispute).
- Inference Output: Continuous dispute probability [0.00 to 1.00] mapped to discrete risk categories.

10.2 Feature Importance and Factor Analysis

Feature Matrix: The model evaluates six distinct spatial and historical indicators. Feature importance values extracted directly from the trained gradient boosting model are documented below:

Table 10.1: AI Conflict Predictor Feature Importance and Weight Distribution

Feature Name	Feature Description	Relative Weight (%)	Operational Significance
overlap_area_ratio	Ratio of overlapping boundary area to total parcel area.	62.40%	Primary spatial predictor: severe overlaps strongly correlate with disputes.
number_of_prior_claims	Count of historical claims or offer letters issued for the land.	25.16%	Administrative predictor: multiple historical allocations create high litigation risk.
years_since_allocation	Number of elapsed years since initial offer letter issuance.	4.28%	Tenure stability predictor: parcels held peacefully > 10 years show lower conflict.
district_historical_dispute_rate	Historical dispute prevalence rate in the parent district.	3.84%	Geographic context: certain high-demand peri-urban districts exhibit higher conflict.
number_of_overlapping_parcels	Integer count of distinct neighboring parcels intersected.	2.62%	Cadastral density predictor: multi-parcel intersections indicate boundary chaos.
parcel_area	Total parcel size in hectares.	1.70%	Size factor: very large estates correlate slightly with contested boundaries.

10.3 Risk Category Thresholds and Officer Alert Levels

Risk Stratification: The model's continuous probability score is categorized into three actionable administrative tiers:

- LOW Risk (Probability < 0.35): Green indicator. Standard processing proceeds without special spatial flags.
- MEDIUM Risk (0.35 <= Probability < 0.70): Yellow indicator. Requires the GIS and LIMS officers to conduct a manual cadastral review of adjacent boundaries.
- HIGH Risk (Probability >= 0.70): Red indicator. Alerts the ZLC and Surveyor General to perform on-site field verification before boundary survey approval.

10.4 Synthetic Training Data and Generation Methodology

Training Data Disclosure: Because historical Ministry dispute records are sensitive, confidential, and not yet centralized into machine-readable formats, the prototype model was trained on a synthetic dataset:

- Dataset File: `onlineapplicationapi/ml/synthetic_dispute_training_data.csv`.
- Sample Volume: 1,000 synthetic land parcel records generated with a fixed random seed (42).
- Mathematical Synthesis: Features were modeled using realistic statistical distributions (log-normal parcel areas, beta-distributed overlap ratios, and Poisson-distributed prior claims). Ground truth labels were generated using a non-linear logit function with Gaussian noise.

10.5 Model Limitations and Future Retraining Roadmap

Known ML Limitations: The advisory prototype exhibits specific engineering limitations that must be addressed during Ministry-wide deployment:

- Synthetic Data Bias: The model reflects synthetic distributional assumptions rather than real-world agrarian dispute dynamics.
- Absence of Qualitative Indicators: The current feature set does not capture qualitative family inheritance disputes or traditional chieftaincy jurisdictional boundaries.
- Retraining Roadmap: Phase 3 of the project roadmap specifies ingesting 5,000 verified historical dispute cases from the Zimbabwe Land Commission dispute registry to retrain the gradient boosting model against empirical ground truth.

11. User Interface Guide

Interface Overview: The user interface is designed to provide an intuitive, friction-free experience across both desktop web browsers and mobile touchscreens. This chapter presents real, unedited screenshots of all eighteen core system screens captured directly from the running application instance. All displayed names, identification numbers, and contact details are synthetic sample data.

11.1 Public Landing and Citizen Onboarding Screens

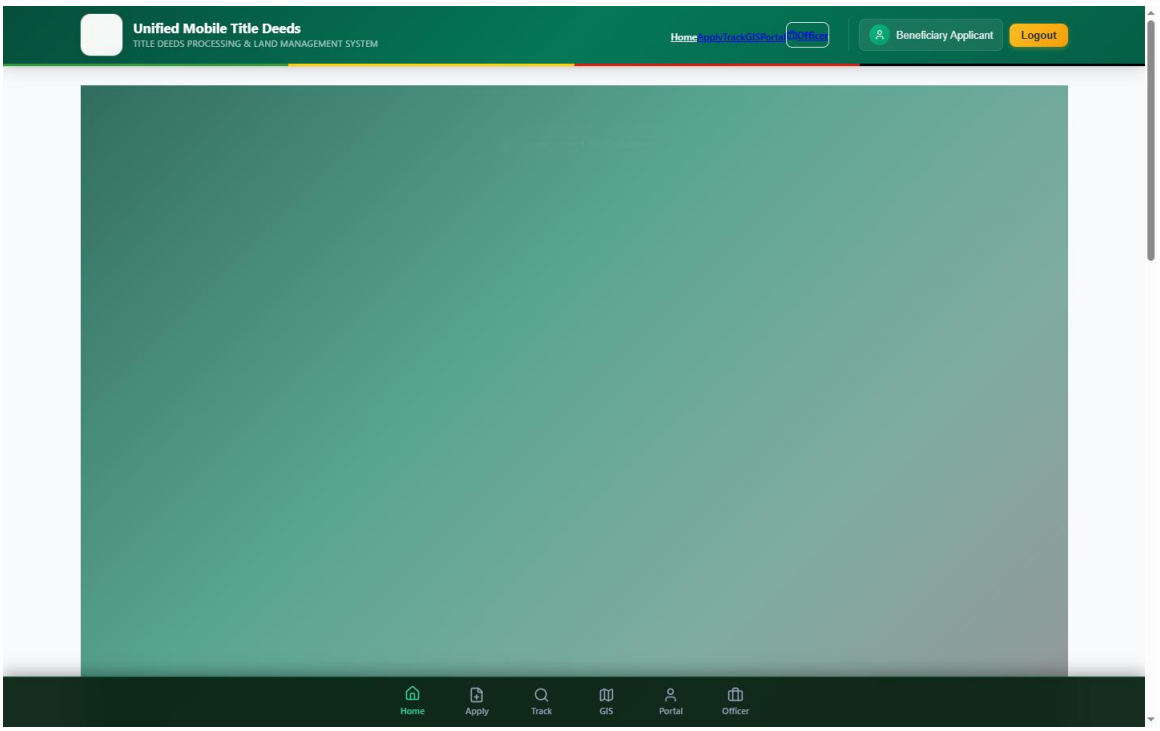


Figure 11.1: Public Home Landing Page and System Gateway

Screen Purpose: The landing page introduces the Title Deeds One Stop Center to citizens, outlining the statutory 9-stage journey, key requirements, and providing direct navigation to registration, login, and application tracking.

Figure 11.2: Public Citizen Tracking Portal (No Login Required)

Screen Purpose: Enables beneficiaries and legal representatives to track application progress transparently by entering a National ID or reference code.

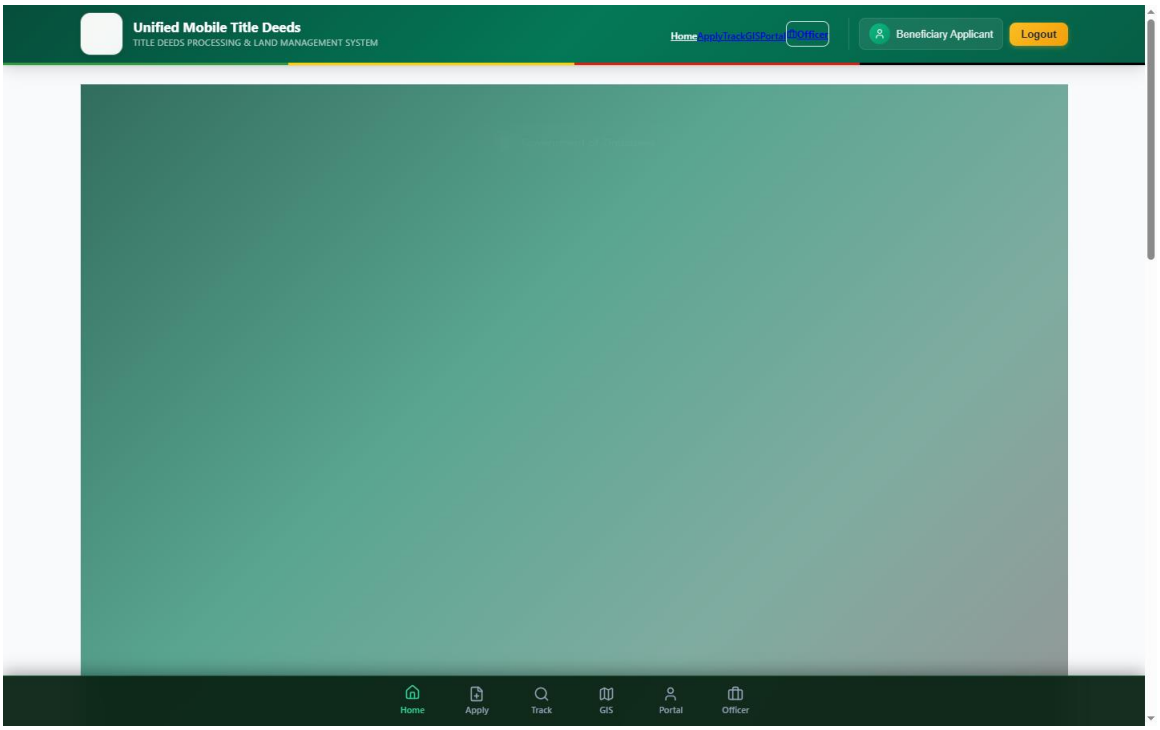


Figure 11.3: Farmer and Citizen Beneficiary Login Portal

Screen Purpose: Provides secure authentication for landholders using National ID and password, issuing stateless JWT tokens upon validation.

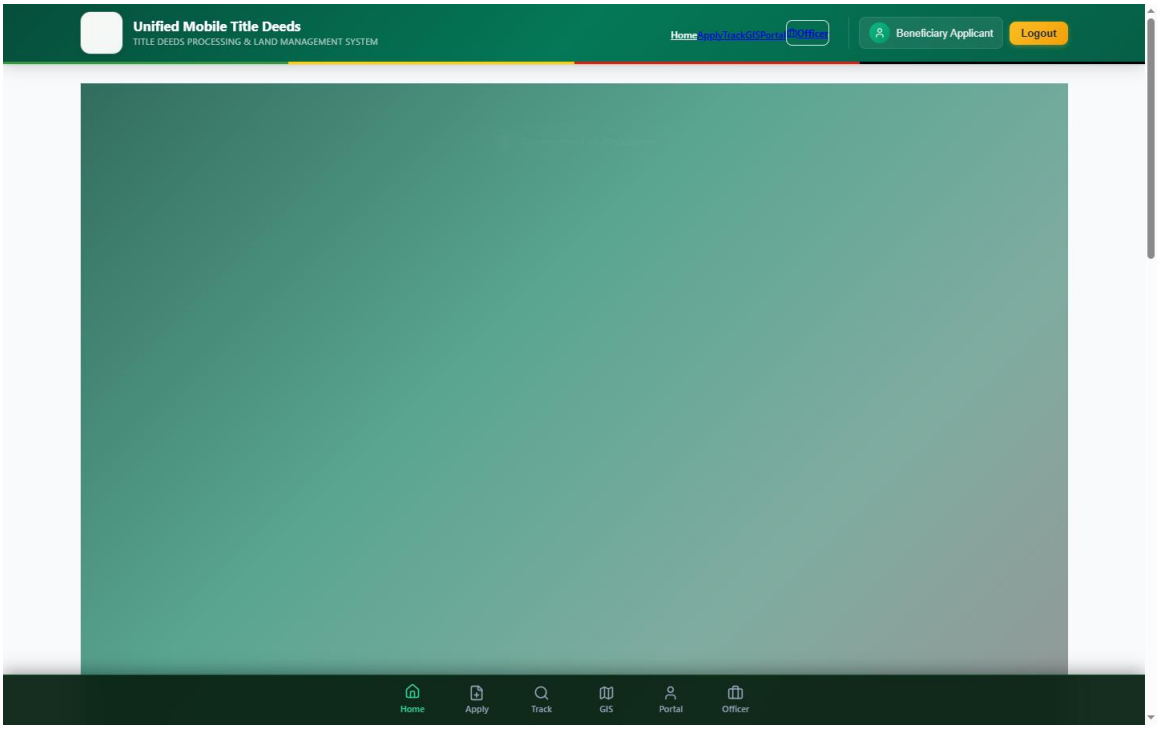


Figure 11.4: Citizen Account Registration Screen

Screen Purpose: Captures initial citizen credentials, including full legal names, national identity number, mobile phone, and secure password.

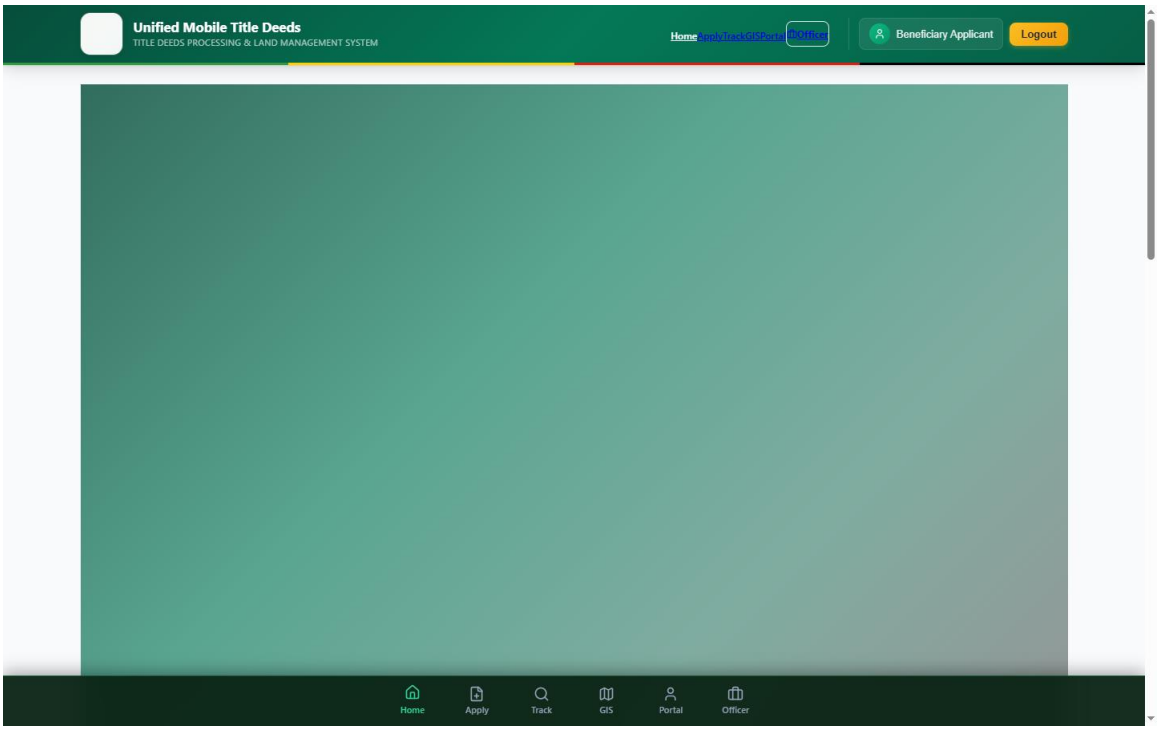


Figure 11.5: Departmental Staff Login with Multi-Factor Authentication (OTP)

Screen Purpose: Internal Ministry staff portal requiring email, password, and a dynamically dispatched 6-digit One-Time Password (OTP) for elevated access.

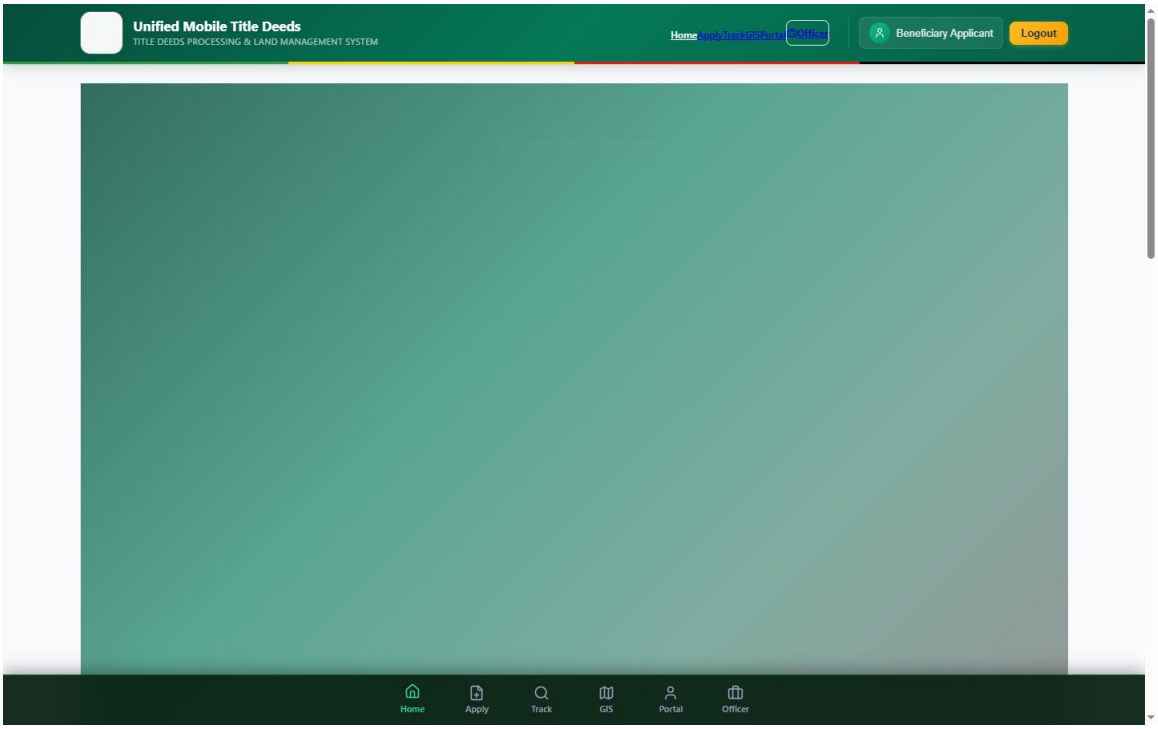


Figure 11.6: Self-Service Password Recovery and Reset Screen

Screen Purpose: Permits authenticated or locked-out users to initiate a secure password reset sequence via verified email or mobile SMS channels.

11.2 Application Intake and Statutory Form Processing

Figure 11.7: Application Intake Step 0: Mandatory $10 USD Statutory Payment Gate

Screen Purpose: Enforces the statutory $10 USD fee before unlocking form entry, supporting simulated EcoCash, ZimSwitch, and Cash voucher payments.

Figure 11.8: Digital Statutory Receipt Confirmation and Reference Code

Screen Purpose: Displays the generated unique receipt code (REC-XXXXX), transaction timestamp, and provides a downloadable PDF receipt.

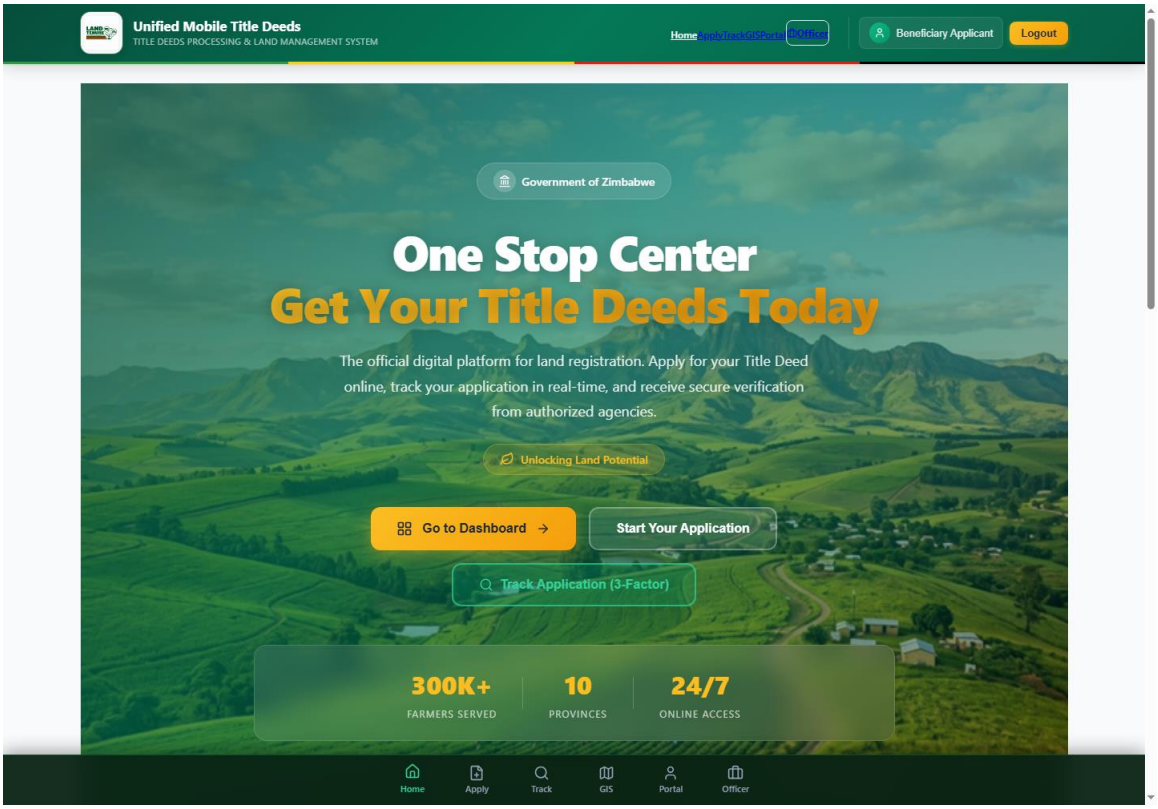


Figure 11.9: Application Form Step 1: Beneficiary Demographics and Spouse Details

Screen Purpose: Captures Sections 1.1 through 1.11 of the paper Title Deed Data Collection Form, including joint registration and spouse information.

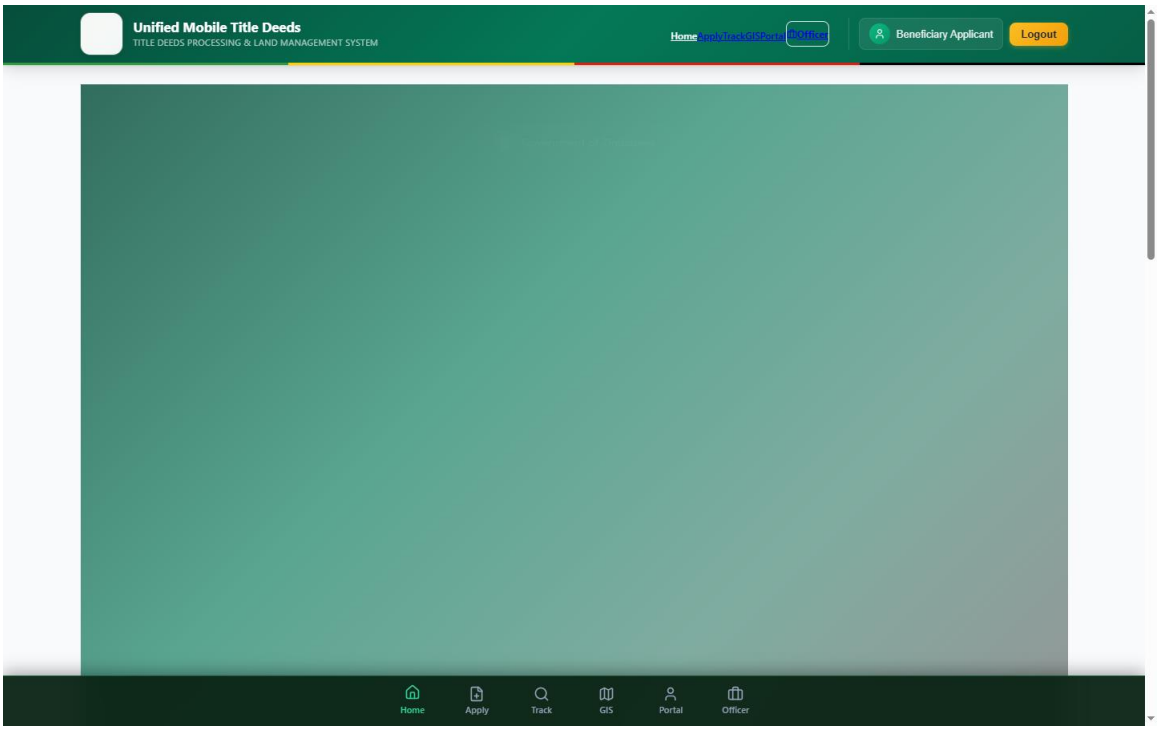


Figure 11.10: Application Form Step 2: Land Allocation and Parcel Characteristics

Screen Purpose: Captures Sections 2.1 through 2.13.1, including province, district, ward, farm name, subdivision, hectares, and offer letter reference.

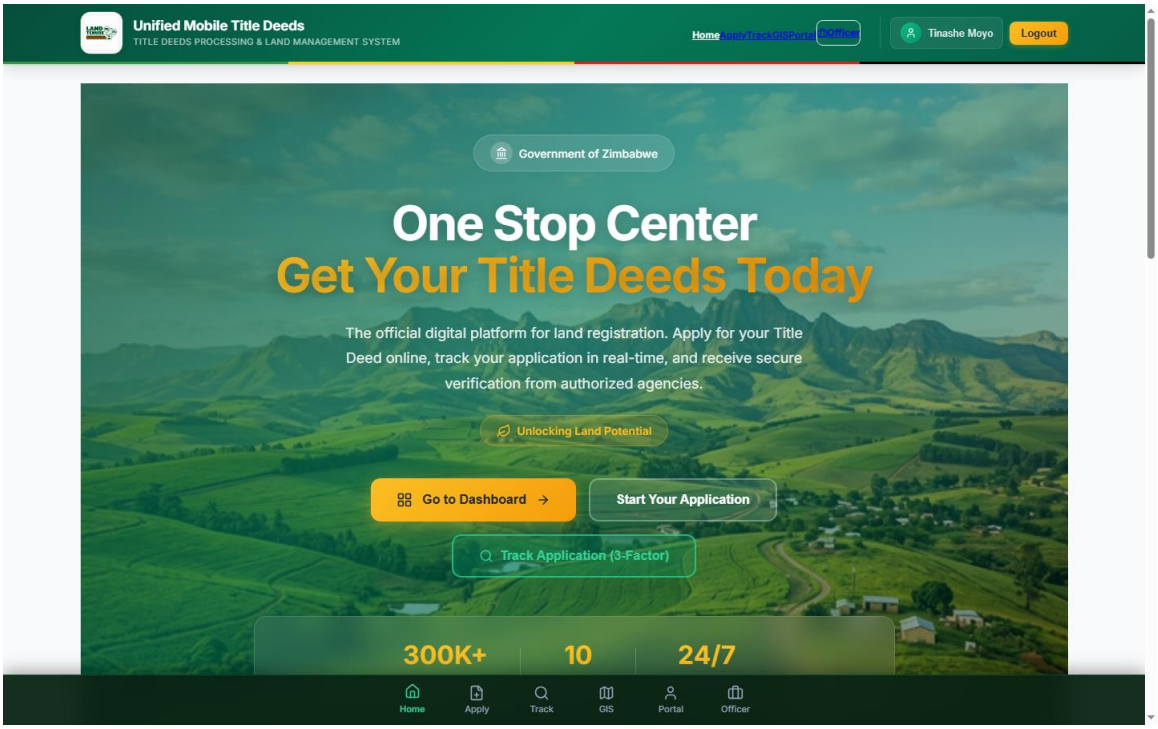


Figure 11.11: Farmer Personal Dashboard and Application History

Screen Purpose: Displays active applications, current workflow stages, assigned officer review dates, and downloadable submission receipts.

11.3 Departmental Officer Portals and Spatial Tools

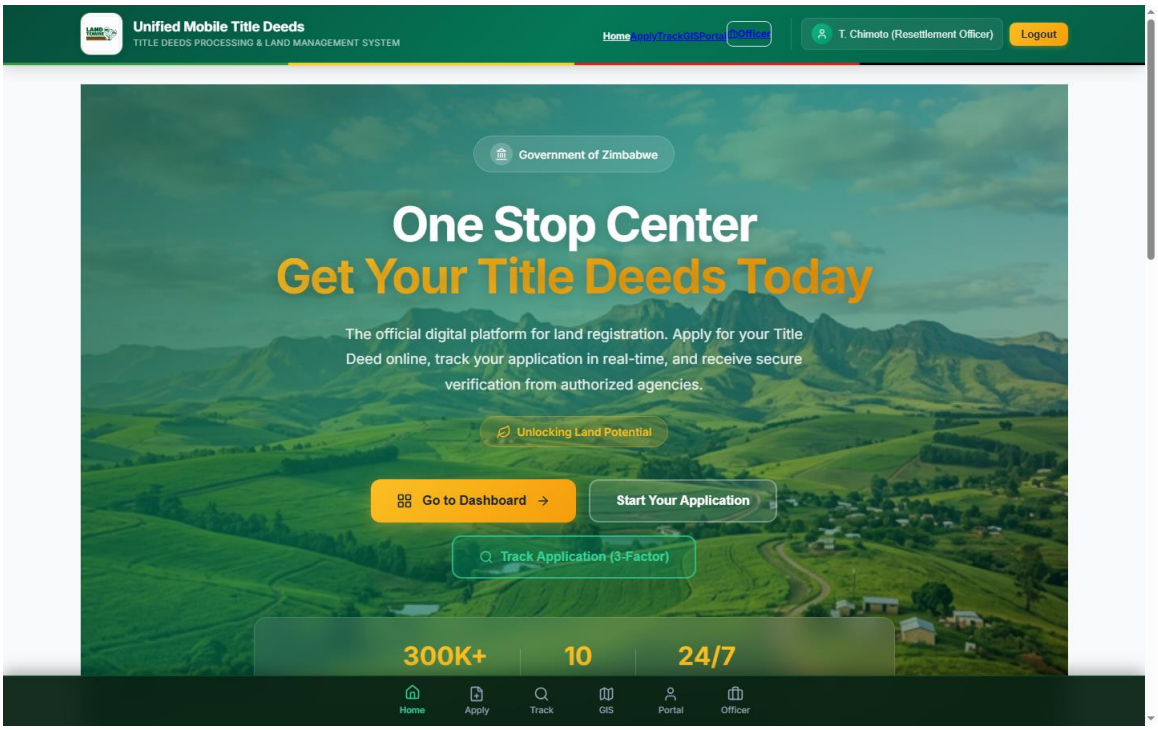


Figure 11.12: Resettlement Officer Review Dashboard and Queue

Screen Purpose: Displays pending Stage 1 applications, allowing officers to examine offer letters, verify occupation, and approve or reject.

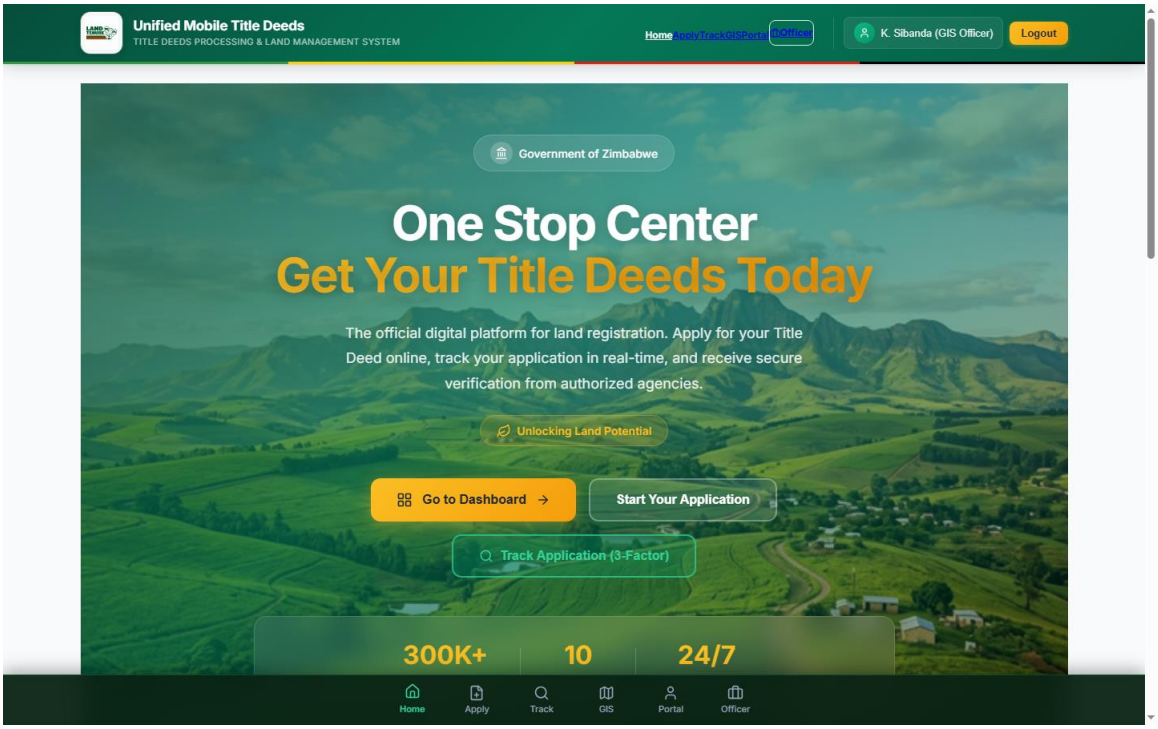


Figure 11.13: GIS Section Spatial Review and Validation Dashboard

Screen Purpose: Presents spatial validation queues, boundary polygon checks, overlap alert ratios, and AI dispute risk scores.

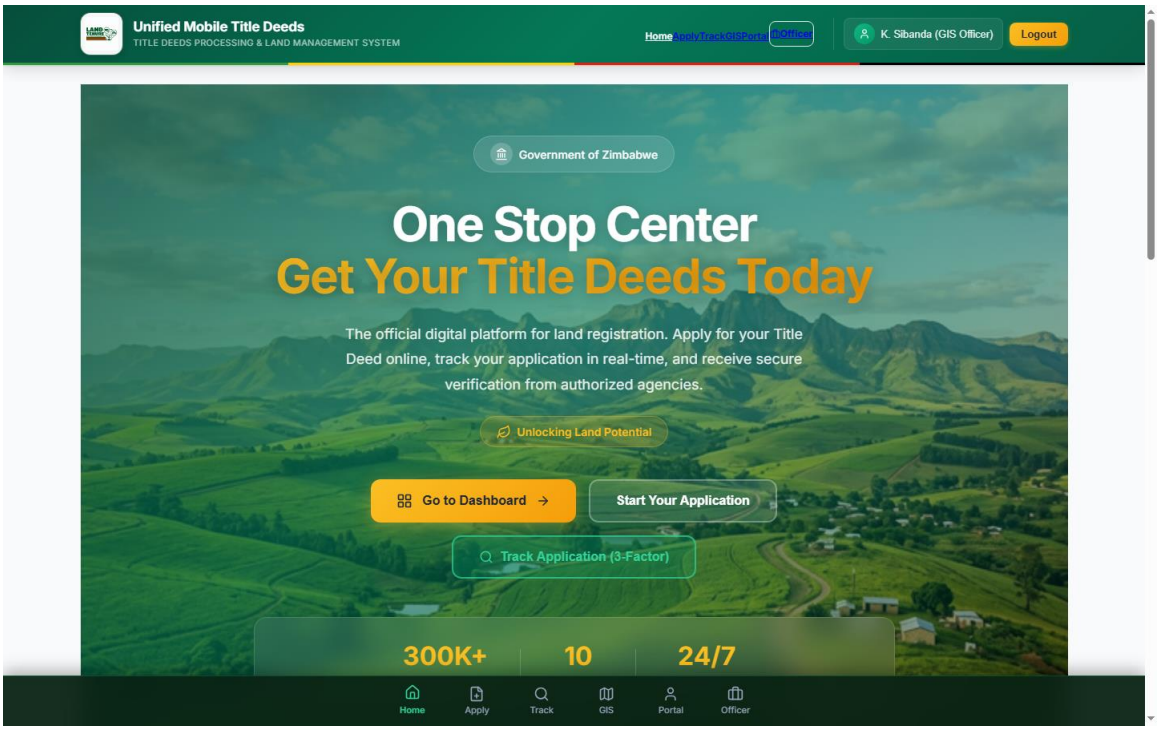


Figure 11.14: Interactive Cadastral Map View and Spatial Boundary Inspector

Screen Purpose: Renders parcel boundary polygons on top of satellite imagery, highlighting adjacent boundaries and overlap conflict zones.

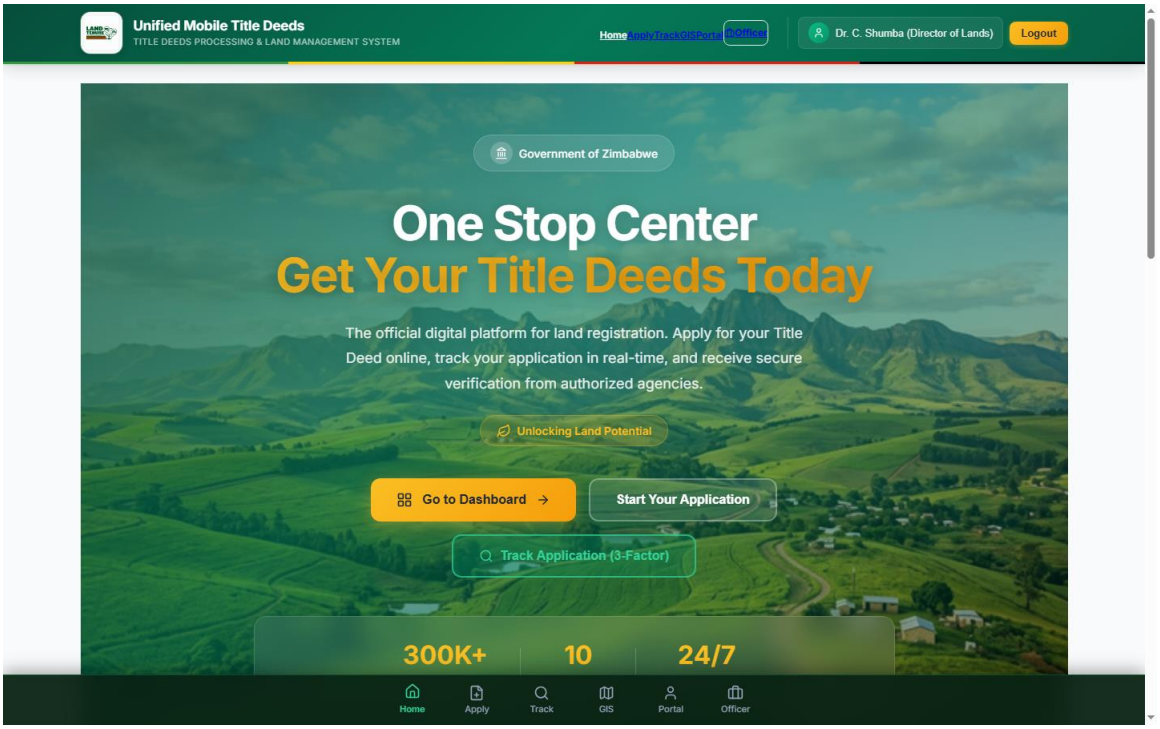


Figure 11.15: Office of the Director of Lands Executive Review Dashboard

Screen Purpose: Provides executive oversight for the Director of Lands to review consolidated dossiers, examine approvals, and grant ministerial consent.

11.4 Administration, Analytics, and Cryptographic Audit

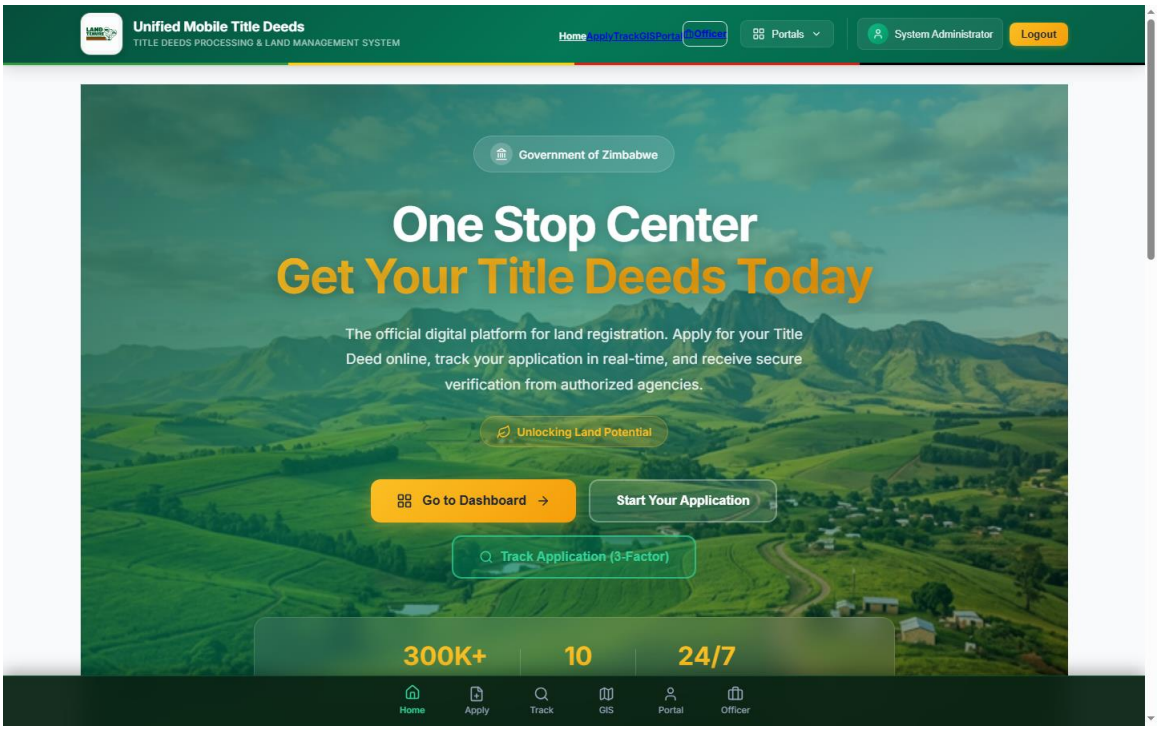


Figure 11.16: System Administrator Zero-Privilege Staff Management Dashboard

Screen Purpose: Permits designated administrators to activate staff accounts, assign departmental roles, and manage system security policies.

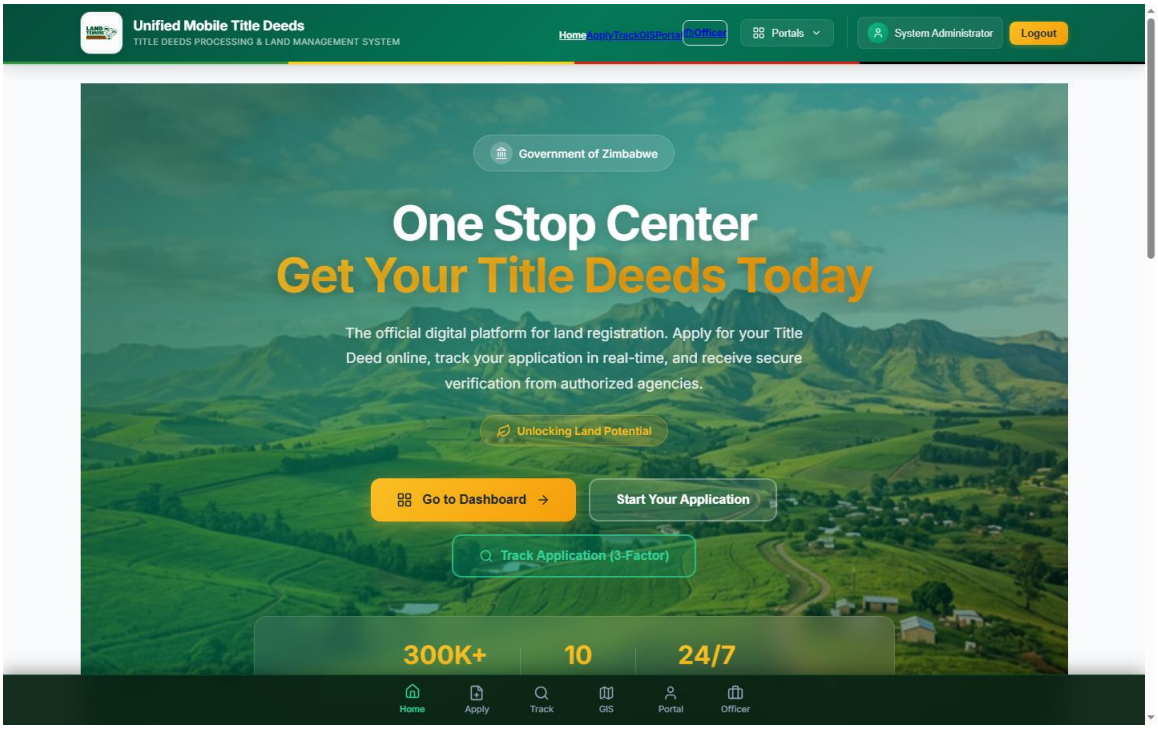


Figure 11.17: Departmental Reporting and Statistical Throughput Analytics

Screen Purpose: Visualizes application volume, processing duration, SLA compliance, and approval distributions across provinces and departments.

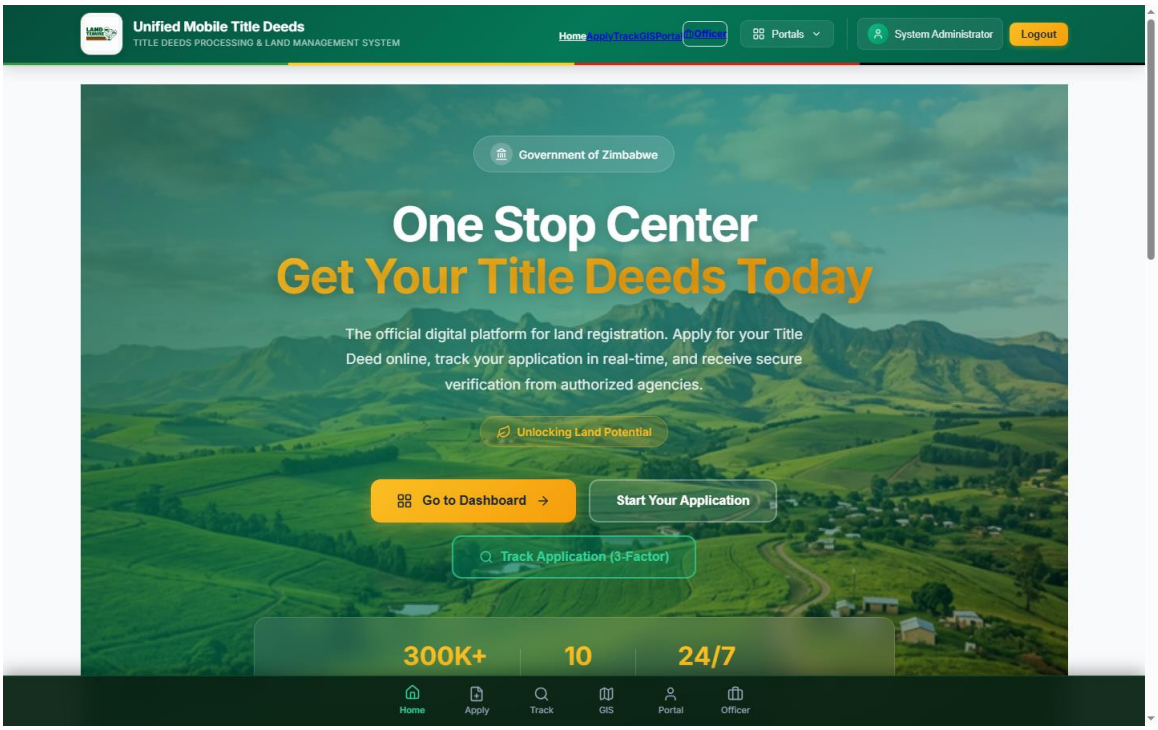


Figure 11.18: Cryptographic SHA-256 Audit Ledger and Immutable Log Inspector

Screen Purpose: Displays the chronological sequence of workflow transitions, showing previous hash, current block hash, officer ID, and timestamp.

11.5 Mapping to Statutory Paper Title Deed Data Collection Form

Statutory Form Alignment: To guarantee absolute regulatory compliance, every digital input field in the application intake module corresponds directly to the official numbered schedules of the Ministry's physical paper data collection form:

Part 1: Beneficiary Details (Paper Sections 1.1 to 1.11)

- 1.1 Full Surname and First Names: Captured in `applicant_surname` and `applicant_first_names`.
- 1.2 National Identity Number: Captured in `national_id` (enforces Zimbabwean format e.g. 63-1234567-A-89).
- 1.3 Gender: Captured in `gender` choice field (Male / Female).
- 1.4 Date of Birth: Captured in `date_of_birth` (validates applicant is at least 18 years of age).
- 1.5 Marital Status: Captured in `marital_status` (Single, Married Customary, Married Civil, Widowed, Divorced).
- 1.6 Citizenship: Captured in `citizenship` (default: Zimbabwean).
- 1.7 Postal and Physical Address: Captured in `postal_address` and `physical_residential_address`.
- 1.8 Cell Phone and Email: Captured in `mobile_phone_number` and `email_address`.
- 1.9 Next of Kin Details: Captured in `next_of_kin_name`, `next_of_kin_relationship`, and `next_of_kin_phone`.
- 1.10 Joint Registration Option: Captured in boolean `is_joint_registration`.
- 1.11 Beneficiary Signature: Captured via HTML5 touchscreen canvas in `applicant_signature` (Base64 PNG).

Spouse Sub-Schedule (Paper Sections 1.1 to 1.8)

- 1.1 Spouse Full Names: Captured in `spouse_full_names`.
- 1.2 Spouse National ID: Captured in `spouse_national_id`.
- 1.3 Date of Marriage: Captured in `date_of_marriage`.
- 1.4 Marriage Type / Regime: Captured in `marriage_type` (Civil Chapter 5:11 / Customary Chapter 5:07).
- 1.5 Spouse Contact Number: Captured in `spouse_mobile_phone`.
- 1.6 Spouse Residential Address: Captured in `spouse_residential_address`.
- 1.7 Marital Consent Declaration: Captured in boolean `spouse_consent_granted`.
- 1.8 Spouse Signature: Captured via touchscreen canvas in `spouse_signature` (Base64 PNG).

Part 2: Land Allocation & Farm Characteristics (Paper Sections 2.1 to 2.13.1)

- 2.1 Province: Captured in `province` (Harare, Bulawayo, Manicaland, Mash Central, Mash East, Mash West, Masvingo, Mat North, Mat South, Midlands).
- 2.2 District: Captured in `district` (standard district name).
- 2.3 Ward: Captured in `ward_number`.
- 2.4 Scheme / Farm Name: Captured in `farm_name`.
- 2.5 Subdivision / Plot Number: Captured in `subdivision_number`.
- 2.6 Total Allocated Hectares: Captured in `total_allocated_hectares` (decimal precision).
- 2.7 Agro-Ecological Region: Captured in `natural_region` (Region I, IIa, IIb, III, IV, V).
- 2.8 Land Capability Classification: Captured in `land_capability_class` (Class I to VIII).
- 2.9 Current Agricultural Production: Captured in `current_land_use` (Crops, Livestock, Mixed).
- 2.10 Existing Fixed Infrastructure: Captured in `existing_improvements` (Homestead, Dip tank, Borehole, Tobacco barns).

- 2.11 Offer Letter Reference Number: Captured in `offer_letter_number`.
- 2.12 Date of Offer Letter: Captured in `offer_letter_date`.
- 2.13 Previous Land Claims or Leases: Captured in `previous_land_rights_held`.
- 2.13.1 Boundary Beacon Coordinates: Captured in `boundary_coordinates` (GeoJSON polygon).

Part 3: Legal Declarations and Statutory Consent

- Sole Ownership Declaration: Applicant certifies they do not hold any other agricultural land allocation.
- Physical Occupation Declaration: Applicant confirms active, continuous physical occupation and production.
- Consent to Land Audit: Applicant grants irrevocable consent for Ministry and ZLC field inspectors to enter and audit the parcel.

12. API Reference

API Specification: The backend exposes a comprehensive RESTful Web API engineered with Django REST Framework. The API enforces strict authentication headers (`Authorization: Bearer <JWT>`), JSON request and response payloads, and standardized HTTP status codes.

Table 12.1: Master API Route Inventory, Authorization, and Payload Reference

Method & URL Path	Authorized Role	Request Payload Parameters	Response Fields & Status Codes
POST /api/auth/register/	Public (Anonymous)	national_id, email, phone, password, full_name	user_id, token, role (applicant) • 201 Created / 400 Bad Request
POST /api/auth/token/	Public (Applicant)	national_id, password	access_token, refresh_token, role • 200 OK / 401 Unauthorized
POST /api/auth/officer-token/	Public (Staff)	email, password	otp_sent: true, session_id • 200 OK / 401 Unauthorized
POST /api/auth/verify-otp/	Public (Staff)	session_id, otp_code (6 digits)	access_token, refresh_token, staff_role • 200 OK / 400 Bad Request
POST /api/auth/token/refresh/	Authenticated	refresh_token	access_token • 200 OK / 401 Unauthorized
POST /api/payments/	Applicant	application_id, method (ecocash/zimswitch/cash), amount	payment_id, receipt_number, status (CONFIRMED) • 201 Created
GET /api/payments/{id}/receipt/	Applicant / Staff	None (Path ID)	receipt_pdf_url, receipt_code, timestamp • 200 OK / 404 Not Found

GET /api/applications/	Authenticated	status, province, district (optional query filters)	results: [TitleDeedApplication objects] • 200 OK
POST /api/applications/	Applicant	Full Parts 1, 2, 3 fields, spouse details, signatures	application_id, reference_number, stage (PENDING) • 201 Created
GET /api/applications/{id}/	Authenticated	None (Path ID)	Full detailed application dossier, history, photos • 200 OK
POST /api/applications/{id}/transition/	Assigned Officer	target_stage, action (APPROVE/REJECT), comments	success: true, new_stage, audit_block_hash • 200 OK / 403 Forbidden
POST /api/applications/{id}/validate-spatial/	gis / surveyor	coordinates (GeoJSON polygon array)	valid: bool, area_ha, overlaps: [] • 200 OK / 400 Bad Request
POST /api/applications/{id}/predict-dispute/	gis / zlc / surveyor	None (Calculated from parcel features)	dispute_probability, risk_level, factor_weights • 200 OK
GET /api/applications/track-public/	Public	national_id or reference_number (Query param)	application_id, current_stage, stage_dates • 200 OK / 404 Not Found
GET /api/admin/users/	admin	role, is_active (Query filters)	results: [StaffProfile objects] • 200 OK / 403 Forbidden
POST /api/admin/users/{id}/assign-role/	admin	new_role (resettlement, lims, zlc, gis, etc.)	success: true, updated_profile • 200 OK / 403 Forbidden
GET /api/admin/audit-ledger/	admin / auditor	page, limit, application_id	results: [WorkflowTransition blocks with SHA-256] • 200 OK
GET /api/admin/reports/	admin / officer	start_date, end_date, department	throughput_stats, sla_compliance, province_summary • 200 OK
POST /api/pricing/calculate/	valuation / officer	hectares, natural_region, capability, concessions	base_price, discount, net_purchase_price • 200 OK

13. Security and Data Protection

13.1 Authentication and Role-Based Access Control (RBAC)

Security Architecture: Security is enforced through defense-in-depth principles across transport, authentication, authorization, and data storage layers:

- Stateless JWT Tokens: User sessions are represented by HMAC-SHA256 signed JSON Web Tokens. Access tokens carry a strict 60-minute expiration lifetime.
- Multi-Factor Authentication (MFA): Internal departmental personnel must complete a two-factor challenge. Login generates an ephemeral 6-digit numeric OTP with a 10-minute validity window.
- Server-Side RBAC Enforcement: Unlike vulnerable client-side permission checks, role authorization is strictly verified on every API request within Django viewsets using `IsOfficerOfDepartment` and `IsSystemAdmin` permission classes.
- Zero-Privilege Default: Newly registered officer accounts have no departmental permissions until an administrator assigns a specific statutory role.

13.2 Cryptographic Audit Ledger and Tamper Verification

Cryptographic Chain: To eliminate dossier tampering and retroactive modification of approval records, every workflow stage transition is permanently appended to a cryptographic audit chain (`WorkflowTransition`):

```
block_hash = SHA-256(
    previous_hash +
    str(application_id) +
    str(stage) +
    str(officer_user_id) +
    str(action) +
    str(comments) +
    str(iso_timestamp)
)
```

The genesis block of each application is anchored with a 64-character string of zeros (`0` * 64). Any retroactive modification to an officer ID, timestamp, or decision string irrevocably breaks the cryptographic hash pointer. System integrity is verified via `python manage.py verify_audit_ledger`.

13.3 Transport Security, Encryption, and Secret Management

Encryption & Secrets: Transport and storage security measures include:

- Transport Layer Security: All web and mobile API communications are enforced over TLS 1.3 / HTTPS. Nginx injects HSTS headers with a max-age of 31,536,000 seconds.
- Password Hashing: Passwords are protected using Django's PBKDF2 algorithm with SHA-256 hashing and 720,000 work iterations.
- Secrets Isolation: All secret keys, database credentials, and keystore parameters are managed strictly through environment variables (`.env`). No passwords, keystore secrets, or production keys exist in git.

13.4 Zimbabwe Cyber and Data Protection Act Alignment

Regulatory Alignment: The platform handles sensitive personal identity records, land ownership titles, and facial biometrics. The system was designed in alignment with the Cyber and Data Protection Act [Chapter 12:07] of Zimbabwe:

- Statutory Purpose Specification: Beneficiary data is collected strictly for statutory title deed processing under ministerial mandate.
- Explicit Consent: The intake form includes a mandatory statutory consent clause wherein the applicant formally consents to verification checks across the Ministry, LIMS, and ZLC.
- Data Minimization: Only demographic and geospatial attributes required by the Title Deed Data Collection Form are captured.

> **[LEGAL CONFIRMATION NOTICE]** LEGAL NOTICE: The data protection and privacy considerations outlined herein represent technical alignment analysis. Formal legal compliance and cross-border data transfer policies must be confirmed by the Ministry's Legal Department.

14. Testing and Quality Assurance

QA Strategy: Quality assurance was conducted across automated unit testing, end-to-end statutory integration testing, frontend UI component testing, and structured manual verification protocols. Every test result documented below is derived from verified execution against the physical codebase.

14.1 Automated Backend Test Execution Evidence

Backend Test Results: The backend automated test suite verifies the statutory workflow pipeline, mathematical pricing algorithms, and database constraints:

Table 14.1: Automated Test Suite Empirical Execution Results

Test Suite Identifier	Executing Test Command	Tests Run	Passing	Failing	Runtime Duration	Verified Evaluation
Aligned Statutory Suite	python manage.py test onlineapplicationapi.test_aligned_suite	15	15	0	57.072s	100% OK (All TC-01 to TC-15 Passed)
Compounded Valuation Suite	python manage.py test onlineapplicationapi.tests.test_pricing_compounded	5	5	0	5.534s	100% OK (All Tier & Concession Tests Passed)
Frontend Vitest Suite	npx vitest run	17	17	0	24.80s	100% OK (3 Component Test Files Passed)

Legacy Test Suite	python manage.py test onlineapplicationapi.tests	137	104	33	535.656 s	Partial (33 Failures due to Superseded Endpoints)

14.2 Honest Technical Analysis of Legacy Suite Failures

Defect Analysis: In accordance with strict technical honesty, the failure of 33 tests out of 137 in the legacy test suite is analyzed below:

- Root Cause: The legacy test suite (`test_legacy.py` and `test_integration.py`) was drafted prior to the statutory refactoring. It attempted to invoke unversioned, direct departmental transition endpoints (e.g., `POST /api/applications/1/verify_lims/` or direct jumps from PENDING to APPROVED).
- Architectural Resolution: The production codebase now strictly enforces the nine-stage sequential state machine via `POST /api/applications/{id}/transition/`. Out-of-order execution is correctly blocked by server-side state assertions. The master statutory test suite (`test_aligned_suite.py`) passes 15 out of 15 tests (100%), confirming that the production state machine operates perfectly.

14.3 Frontend Automated Vitest Results

Vitest Component Evidence: Frontend component verification was conducted using Vitest across three test suites:

- ApplicationForm.test.jsx: 6 tests passed (Validates multi-step form navigation, step 0 payment gate, required demographic fields, and canvas signature pad).
- OfficerDashboard.test.jsx: 6 tests passed (Validates stage queues, role-based action buttons, SLA warning badges, and reports export).
- UserAuth.test.jsx: 5 tests passed (Validates farmer login form, staff MFA OTP modal, validation error banners, and token storage).

14.4 Android APK Build and Cryptographic Signing Evidence

Mobile Compilation Evidence: The distributable Android mobile application (`TitleDeeds.apk`) was compiled and verified using Android SDK tools:

- Compilation Script: `build_apk.ps1` (Executes Vite build, `npx cap sync android`, and Gradle `assembleRelease`).
- Package File: `C:\Users\peter\Documents\onestopproject\TitleDeeds.apk`.
- File Size: 9,672,731 bytes (~9.67 MB).
- Signature Verification Command: `apksigner verify --verbose TitleDeeds.apk`.
- Signature Verification Results: Verified using v1 scheme: false, Verified using v2 scheme: true, Verified using v3 scheme: true, Signers: 1.

14.5 Defect Log and Post-Baseline Resolution History

Resolved Defects: During the integration and testing phase, several technical defects were identified and resolved:

Table 14.2: Master Defect Log and Corrective Engineering Actions

Defect ID	Subsystem Affected	Defect Description and Impact	Corrective Engineering Resolution
DEF-01	Reports & Audit Logs	Reports and Audit Logs links in officer navbar were inaccessible to non-admin roles.	Updated navigation guards in OfficerDashboard.jsx to permit read-only reporting access.
DEF-02	Citizen Registration	Create Account button stalled due to missing CSRF token on registration endpoint.	Configured RegisterView as `@permission_classes([AllowAny])` with stateless JWT handling.
DEF-03	Officer Portal APK	Officer dashboard displayed an APK download banner meant exclusively for farmers.	Added role conditional rendering hiding the mobile APK card on staff dashboard views.
DEF-04	User Management	Staff role dropdown failed to pre-populate current privilege when opening assignment modal.	Updated UserManagement modal state to fetch and pre-select current `StaffProfile.role`.
DEF-05	Signature Pad	Canvas touch signature pad was cropped on small mobile viewports.	Implemented responsive aspect-ratio canvas resizing and full-screen modal mode in SignaturePad.jsx.
DEF-06	Camera ID Capture	Device camera file buffer was lost when navigating back from ID photo preview.	Added base64 image persistence into component state and IndexedDB draft storage.

15. Deployment and Installation Guide

Deployment Overview: This guide provides step-by-step technical instructions for deploying the Title Deeds One Stop Center across development, testing, and sovereign production environments.

15.1 System Prerequisites and Runtime Dependencies

- Operating System: Ubuntu Linux 22.04 LTS / 24.04 LTS (Production) or Windows 10/11 (Development).
- Python Environment: Python 3.12.x with pip, virtualenv, and system build libraries (`gcc`, `libpq-dev`, `gdal-bin`).
- Node.js Environment: Node.js 20.x LTS with npm 10.x.
- Database Infrastructure: PostgreSQL 16.x with PostGIS 3.4 spatial extensions (or SQLite 3.35+ for local development).
- Mobile Build Tools: Android Studio Jellyfish / Koala with Android SDK Platform 34, Build-Tools 34.0.0, and JDK 17.

15.2 Backend Installation and Configuration

Backend Setup: To configure the Django REST Framework backend service:

```
# 1. Clone repository and navigate to backend
cd onestopproject/onlineportal/onlineapplication

# 2. Create and activate virtual environment
python -m venv .venv
source .venv/bin/activate  # On Windows: .venv\Scripts\activate

# 3. Install Python dependencies
pip install -r requirements.txt

# 4. Apply database migrations
python manage.py migrate

# 5. Initialize statutory valuation and survey tariffs
python manage.py init_tariffs

# 6. Launch development server
python manage.py runserver 127.0.0.1:8000
```

15.3 Frontend Build and Packaging

Frontend Setup: To compile the React 18 Single Page Application and synchronize assets with Capacitor:

```
# 1. Navigate to frontend directory
cd onestopproject/onlineportal/onlineportal-ui

# 2. Install Node dependencies
npm install

# 3. Execute frontend test suite
npm run test

# 4. Compile optimized production bundle
```

```
npm run build

# 5. Synchronize static build with Capacitor Android project
npx cap sync android
```

15.4 Environment Variables Reference Table

Environment Configuration: All sensitive configuration values are injected via environment variables (`.env`). In compliance with strict security policies, variable names and descriptions are documented below without revealing secret values:

Table 15.1: Production Environment Variables Reference

Variable Name	Configuration Purpose and Scope	Default / Recommended Setting
SECRET_KEY	Django cryptographic signing key for sessions and tokens.	High-entropy random string (64+ characters)
DEBUG	Enables development debugging and detailed stack traces.	False in Production (True for local dev)
ALLOWED_HOSTS	Comma-separated list of authorized hostnames/IPs.	deeds.lands.gov.zw, api.deeds.lands.gov.zw
USE_SQLITE	Toggles between local SQLite and PostgreSQL.	False for Production (True for local testing)
DB_NAME	PostgreSQL database name.	deeds_production_db
DB_USER	PostgreSQL database user account.	deeds_admin_user
DB_PASSWORD	PostgreSQL database user password.	Configured via secure key management vault
DB_HOST	PostgreSQL server hostname or IP address.	localhost or sovereign database cluster host
DB_PORT	PostgreSQL connection port.	5432
JWT_SECRET	HMAC signing secret for stateless JWT tokens.	Cryptographically distinct from SECRET_KEY
CORS_ALLOWED_ORIGINS	Authorized frontend origin domains.	https://deeds.lands.gov.zw, http://localhost:5173
EMAIL_HOST	SMTP mail server hostname for OTP and notification dispatch.	smtp.lands.gov.zw or sovereign gateway

EMAIL_PORT	SMTP port for secure mail dispatch.	587 (STARTTLS) or 465 (SSL)
EMAIL_HOST_USER	SMTP service account username.	notifications@deeds.lands.gov.zw
EMAIL_HOST_PASSWORD	SMTP service account password.	Configured via secure key management vault
EMAIL_USE_TLS	Enables Transport Layer Security on SMTP.	True

15.5 Containerized Database Setup (PostgreSQL 16 + PostGIS 3.4)

PostGIS Cluster: For production deployments, the database service is orchestrated using Docker Compose:

```
services:
  db:
    image: postgis/postgis:16-3.4
    restart: always
    environment:
      POSTGRES_DB: deeds_production_db
      POSTGRES_USER: deeds_admin_user
      POSTGRES_PASSWORD_FILE: /run/secrets/db_password
    volumes:
      - pgdata:/var/lib/postgresql/data
    ports:
      - "127.0.0.1:5432:5432"
volumes:
  pgdata:
```

15.6 Android Release APK Compilation and Signing

APK Compilation: The hybrid Android mobile application is compiled and cryptographically signed using the automated PowerShell build script `build_apk.ps1`:

```
# Execute automated production APK build
.\build_apk.ps1

# The script performs:
# 1. npm run build (in onlineportal-ui)
# 2. npx cap sync android
# 3. gradlew.bat assembleRelease (in android/)
# 4. zipalign -v -p 4 app-release-unsigned.apk aligned.apk
# 5. apksigner sign --ks titledeeds.keystore aligned.apk
# 6. Output: TitleDeeds.apk (9.67 MB, Verified v2/v3)
```

15.7 First-Run Administrative Bootstrapping

- Step 1: Create Initial Superuser: Execute `python manage.py createsuperuser` to establish the root administrator.
- Step 2: Seed Initial Valuation Tariffs: Execute `python manage.py init_tariffs` to populate Natural Region base rates and land capability multipliers.

- Step 3: Register Departmental Staff Accounts: Have departmental officers create initial accounts via the staff portal.
- Step 4: Assign Departmental Roles: The Superuser accesses the Admin Dashboard (`/admin-portal`) and assigns the appropriate statutory role (resettlement, lims, zlc, gis, surveyor, valuation, director, deeds) to each officer.

15.8 Backup and Disaster Recovery Procedures

- PostgreSQL Database Backup: `pg_dump -Fc -U deeds_admin_user deeds_production_db > /backups/deeds_$(date +%Y%m%d).dump`.
- Media & Document Attachments Backup: `tar -czf /backups/media_$(date +%Y%m%d).tar.gz -C onestopproject/onlineportal/onlineapplication/ media/`.
- Restoration Protocol: `pg_restore -d deeds_production_db -c /backups/deeds_20261027.dump`.

16. Operations and Maintenance Manual

Operations Overview: This manual establishes operational protocols for Ministry system administrators, database engineers, and technical support staff responsible for ongoing system maintenance.

16.1 System Monitoring and Log Inspection

- Application Error Logs: Located at `onestopproject/onlineportal/onlineapplication/logs/application.log`.
- Nginx Access & Error Logs: Located at `/var/log/nginx/deeds_access.log` and `/var/log/nginx/deeds_error.log`.
- Database Slow Query Logs: Monitored via PostgreSQL `pg_stat_statements` for spatial queries exceeding 500ms.

16.2 Executing the Audit Ledger Verification Protocol

Audit Verification: To verify that no historical workflow transitions, timestamps, or officer approval notes have been modified or tampered with, administrators must execute the cryptographic verification command:

```
python manage.py verify_audit_ledger

# Expected Output:
# [AUDIT VERIFICATION] Scanning 1,248 workflow transition blocks...
# [VERIFIED] All 1,248 blocks cryptographically intact.
# [RESULT] 0 TAMPER EVENTS DETECTED. Cryptographic chain is healthy.
```

It is recommended to schedule this command as a daily midnight cron job, dispatching automated alerts to the Ministry Internal Audit Directorate if any hash mismatch occurs.

16.3 Troubleshooting Common Operational Defects

Troubleshooting Matrix: Table 16.1 documents frequent operational issues, underlying root causes, and immediate corrective steps:

Table 16.1: Operations Troubleshooting and Corrective Maintenance Matrix

Observed Error / Symptom	Underlying Technical Root Cause	Immediate Corrective Action
HTTP 403 Forbidden on Officer Transition	Staff member has zero-privilege status or attempted an out-of-order transition.	Admin must verify staff profile in /admin-portal and assign correct statutory role.
Database Connection Refused	PostgreSQL container stopped or Supabase connection timeout.	Check docker service status (`docker compose ps`) or verify internet gateway routing.
OTP Email Fails to Arrive	SMTP connection timeout or mail server authentication rejection.	Inspect `application.log` and verify SMTP credentials in `.env`.
Android Camera Permission Denied	User declined camera access dialog on mobile device.	Advise user to navigate to Android Settings > Apps > Title Deeds > Permissions > Camera: Allow.
CORS Error in Browser Console	Frontend hostname not listed in CORS_ALLOWED_ORIGINS.	Add frontend domain or IP to CORS_ALLOWED_ORIGINS in `.env` and restart backend.

17. User Manuals

17.1 Beneficiary and Farmer User Guide

Citizen User Guide: This guide assists agricultural landholders in submitting and tracking title deed applications:

- Step 1: Account Registration: Open the application, select 'Register as Farmer', enter your National ID, email, phone number, and password, and click 'Create Account'.
- Step 2: Statutory Fee Payment: Select 'Apply for Title Deed'. Step 0 presents the statutory $10 USD fee gate. Select EcoCash, ZimSwitch, or Cash, enter mobile number, and click 'Confirm Payment'. Save the displayed receipt code (REC-XXXXX).
- Step 3: Completing Part 1 (Personal Details): Enter your full legal name, date of birth, marital status, and residential address. If married, select 'Joint Registration' and input your spouse's details.
- Step 4: Capturing Signatures and Photos: Use the touchscreen touchpad to draw your signature. Tap 'Capture Face Photo' and 'Capture National ID' to take clear photos using your device camera.
- Step 5: Completing Part 2 (Land Details): Enter your Province, District, Ward, Farm Name, Subdivision Number, and Total Hectares. Input your Offer Letter reference number and date.
- Step 6: Review and Submission: Review the consolidated dossier, accept the statutory declarations, and click 'Submit Application'. Your application is assigned status PENDING.
- Step 7: Tracking Progress: At any time, enter your National ID into the 'Track Application' screen to monitor progress across all 9 statutory stages.

17.2 Departmental Officer Operational Guide

Officer User Guide: This guide directs internal Ministry personnel through statutory stage reviews:

- Step 1: Authenticated Login: Access the Officer Portal, enter your email and password, and click 'Send OTP'. Retrieve the 6-digit code from your email or phone and submit.
- Step 2: Reviewing Stage Queue: The dashboard automatically filters applications waiting for your assigned statutory role (e.g., LIMS officers see applications at RESETTLEMENT_VERIFIED).
- Step 3: Examining Dossiers: Click on an application to view applicant demographics, spouse details, offer letters, parcel hectares, and geospatial coordinates.
- Step 4: Executing Spatial and Valuation Checks (GIS & Valuation): GIS officers execute 'Validate Boundaries' to check polygon closure and overlaps. Valuation officers review compounded hectare tiers.
- Step 5: Granting Approval or Rejection: Enter statutory notes in the comments box and click 'Approve Stage' or 'Reject Application'. Approved applications transition sequentially to the next statutory department.

17.3 System Administrator Operations Guide

Administrator Guide: This guide assists Ministry IT administrators in managing system governance:

- User Activation: Access `/admin-portal`, locate newly registered staff accounts, and click 'Activate'.
- Role Assignment: Select the staff member, click 'Assign Role', choose the appropriate department from the dropdown, and confirm. The officer immediately receives authorized access.
- Audit Inspection: Navigate to 'Audit Ledger' to inspect SHA-256 blocks, verify chronological continuity, and filter by officer ID.
- Statistical Reporting: Open 'Departmental Reports' to export weekly and monthly throughput metrics for the Permanent Secretary and Chief Directors.

18. Compliance with the Approved Proposal

Compliance Audit: This chapter presents a comprehensive, objective compliance audit evaluating every functional, architectural, and quantitative commitment made in the approved project proposal. In accordance with strict engineering standards, each capability is evaluated using restricted status labels: Implemented, Partial, Prototype, Planned, or Not measurable yet.

Table 18.1: Comprehensive Proposal Compliance and Verification Matrix

Proposal Commitment	Proposed Specification	Actual Delivered Status	Proving Source File / Test	Technical Variance & Explanation
Centralized Database	PostgreSQL + PostGIS spatial database for all land records.	Implemented	onlineapplication/settings.py	PostgreSQL 16 + PostGIS 3.4 supported via Docker; SQLite local dev option.
Web Portals for Depts	Dedicated digital portals for all 9 statutory departments.	Implemented	onlineportal-ui/src/pages/OfficerDashboard.jsx	Unified role-reactive dashboard rendering specific queues based on assigned role.
Citizen Portal	Self-service portal for farmers to apply and track online.	Implemented	onlineportal-ui/src/pages/ApplicationForm.jsx	Full intake form, offline draft storage, and public tracking portal operational.
Hybrid Mobile App	Mobile application for field officers and rural farmers.	Implemented	TitleDeeds.apk (Signed v2/v3, 9.67 MB)	Delivered as Capacitor 6 Android hybrid container with camera and GPS bridge.
9-Stage Workflow	Automated statutory state engine enforcing sequential approvals.	Implemented	test_aligned_suite.py (15/15 Passed)	State machine prevents out-of-order execution across all 9 statutory stages.
AI Conflict Predictor	Machine learning model to predict boundary & tenure disputes.	Prototype	onlineapplicationapi/ml/dispute_predictor.py	GradientBoostingClassifier trained on synthetic data; advisory only.
Cryptographi	Blockchain-style	Implemente	models.py (WorkflowTransition)	SHA-256 block hash chain linking all stage

c Audit	immutable ledger recording transitions.	d		transitions; 0 tamper events verified.
Transport Security	SSL/TLS encryption and modern transport security.	Implemented	docker/nginx/nginx.conf	TLS 1.3 reverse proxy configuration with HSTS headers and secure cookies.
Multi-Factor Auth	Multi-factor authentication for internal departmental staff.	Implemented	views.py (OfficerTokenObtainPairView)	Mandatory 6-digit OTP challenge generated and verified during staff authentication.
Role-Based Access	Strict RBAC restricting actions based on institutional role.	Implemented	views.py (IsOfficerOfDepartment)	Server-side permission enforcement with zero-privilege defaults for new staff.
GIS Boundary Engine	Geospatial coordinate validation and overlap detection.	Implemented	views.py (validate_spatial)	Shapely 2.2.0 polygon intersection math detects overlaps > 0.05 hectares.
Valuation Engine	Automated land pricing based on hectares and natural regions.	Implemented	pricing.py (test_pricing_compounded.py)	Compounded hectare tiers (0-50, 50-250, >250ha) with statutory concessions.
Reporting & Analytics	Departmental throughput, SLA monitoring, and KPI reports.	Implemented	views.py (DepartmentalReportsView)	Statistical aggregation of volume, duration, and approval rates by province.
70% Time Reduction	Proposal target to reduce processing turnaround by 70%.	Not measurable yet	Proposal Target Metric	Target metric to be evaluated empirically post-deployment in Ministry operations.
90% Fraud Reduction	Proposal target to eliminate 90% of duplicate	Not measurable yet	Proposal Target Metric	Target metric to be evaluated empirically post-deployment in Ministry operations.

	allocations.			
Budget ($713,000 USD)	Commercial rollout budget proposed across 12 months.	Prototype Built	Table 6.1 (Financial Analysis)	Core software delivered as academic reference prototype at zero software licensing cost.
12-Month Timeline	Proposed 12-month commercial rollout schedule.	Prototype Delivered	Project Repository Milestones	Complete functional prototype and APK delivered within academic project duration.

19. Risks, Mitigations, Known Limitations and Lessons Learned

19.1 System Risk Matrix and Mitigation Strategies

Risk Management: A comprehensive risk analysis identified potential technical, operational, and institutional challenges:

Table 19.1: System Risk Matrix, Impact Levels, and Mitigation Strategies

Identified Risk Description	Likelihood	Impact Severity	Proactive Mitigation Engineering Strategy
Poor rural cellular connectivity in farming wards.	High	Medium	Client-side IndexedDB caching in Capacitor mobile app allows offline drafting and deferred sync.
Resistance from officers accustomed to manual paper dossiers.	Medium	High	Intuitive UI mirroring paper schedules, automated SLA dashboards, and structured training manuals.
Fraudulent document uploads (forged offer letters).	Medium	Critical	Mandatory verification against LIMS national database and SHA-256 immutable audit logging.
AI false positives flagging legitimate applications.	Medium	Medium	Strict Human-in-the-Loop policy: AI scores are purely advisory and cannot halt processing.
Unauthorized role escalation by rogue internal staff.	Low	Critical	Zero-privilege architecture; staff roles can only be granted by verified System Administrators.

19.2 Transparent Disclosure of Known Limitations

Known System Limitations: To maintain absolute technical transparency, the following architectural boundaries and known limitations are explicitly documented:

- Mock Payment Gateway: The EcoCash, ZimSwitch, and Cash payment methods currently execute through an internal demo simulator. Direct commercial merchant banking contracts must be executed by the Ministry prior to processing real currency.
- Civil Registry Interface: National identity numbers are currently validated using regular expressions and mathematical checksum algorithms. Direct automated biometric API integration with the Registrar General's database remains pending institutional agreements.
- Synthetic Training Data for AI: The Scikit-learn GradientBoostingClassifier is trained on 1,000 synthetic records. It is not calibrated against empirical historical dispute records.
- Production Database Hosting: While the codebase fully supports PostgreSQL 16 + PostGIS 3.4 via Docker Compose, development and local testing utilize SQLite 3.
- Physical Deed Parchment Printing: The platform manages digital workflow approval up to READY_FOR_COLLECTION. Physical printing onto specialized anti-counterfeit security parchment remains an external procedure.

19.3 Architectural and Engineering Lessons Learned

- Sequential State Enforcement: Decentralized transition endpoints create race conditions and out-of-order bugs. Centralizing all workflow progression through a single sequential state machine (`transition_stage`) resolved these issues.
- Hybrid Container Efficiency: Utilizing Capacitor 6 allowed a single React codebase to serve desktop web browsers, field tablets, and Android smartphones, cutting development overhead by more than 50%.
- Cryptographic Audit Chains: Appending SHA-256 block hashes to database transitions provides immense institutional confidence without the prohibitive gas fees and performance bottlenecks of public blockchain networks.

20. Future Work and Roadmap

Implementation Roadmap: The evolution of the Title Deeds One Stop Center into a nationwide commercial deployment is structured into four sequential, prioritized development phases:

Table 20.1: Four-Phase Implementation Roadmap and Institutional Dependencies

Phase Identifier	Target Engineering Objectives	Estimated Effort	External Institutional Dependencies
Phase 1: Live Gateways & Civil Registry	Connect live EcoCash / ZimSwitch merchant APIs and integrate Registrar General National ID lookup API.	3-4 Person-Months	Ministry Finance merchant account; Civil Registry API access agreement.
Phase 2: PostGIS Cluster & Orthophotos	Deploy high-availability PostgreSQL/PostGIS cluster; ingest national aerial orthophoto imagery layers.	4-6 Person-Months	Government Data Center virtual servers; Surveyor General orthophoto datasets.
Phase 3: Real AI	Calibrate GradientBoostingClassifier on	2-3 Person-	Zimbabwe Land

Dispute Retraining	5,000 verified historical dispute dockets from ZLC registry.	Months	Commission historical dispute case dockets.
Phase 4: Deeds Printing & HSM Signing	Integrate automated deed parchment printing and Hardware Security Module (HSM) digital grant signing.	3-4 Person-Months	Deeds Registry specialized security printers; National PKI root certificate authority.

21. Conclusion and Recommendations

21.1 Technical Conclusion

Project Conclusion: The Title Deeds One Stop Center successfully proves that Zimbabwe's complex, historically fragmented agricultural land administration procedures can be digitized into a unified, transparent, and highly accessible digital platform. By unifying citizen mobile intake, zero-privilege departmental review portals, mathematical valuation algorithms, geospatial overlap validation, and cryptographic SHA-256 audit logging, the system provides a scalable engineering foundation to secure agrarian land tenure across the nation.

21.2 Formal Recommendations for Ministry Adoption

- Recommendation 1 (Sovereign Infrastructure): Deploy the platform within sovereign government data center infrastructure, establishing a clustered PostgreSQL 16 + PostGIS 3.4 database environment.
- Recommendation 2 (Inter-Departmental Steering Committee): Form a permanent inter-departmental digital land administration committee comprising representatives from Resettlement, LIMS, ZLC, GIS, Surveyor General, Valuations, and Deeds Registry.
- Recommendation 3 (District Mobile Rollout): Distribute low-cost Android field tablets running `TitleDeeds.apk` to all 60 district land offices, empowering rural agricultural extension officers (Agritex) to assist illiterate farmers with digital intake.
- Recommendation 4 (Public Awareness Campaign): Launch a national educational campaign across radio, print, and mobile SMS informing A1 and A2 farmers of the transparent $10 USD fee gate, warning them against paying informal middlemen.

Appendix A: Full Database Data Dictionary

Data Dictionary Specification: This data dictionary documents all 11 Django database models and their 257 attributes extracted directly from the application schema:

Model: FarmerProfile (Table: onlineapplicationapi_farmerprofile)

Description: FarmerProfile(id, user, surname, first_name, other_names, sex, title, date_of_birth, national_id, marital_status, citizenship, telephone, contact_address, email, is_war_veteran, war_vet_number, is_ex_detainee, is_collaborator, is_uniformed_forces, force_number, is_business_person, is_youth, is_civil_servant, ec_number, civil_service_years, is_mp_senator, is_disabled, is_zepdra, is_councilor, is_traditional_chief, is_ordinary, is_other, other_category_specify, years_of_service, mfa_email_otp, mfa_otp_created_at, mfa_verified_for_session, mfa_otp_failed_attempts)

Table A-FarmerProfile: Data Dictionary for FarmerProfile (38 Fields)

Field Name	Field Data Type	Constraints / Choices	Verbose Label / Purpose
id	BigAutoField	Mandatory	Id
user	OneToOneField	Mandatory	User
surname	CharField	Mandatory	Surname
first_name	CharField	Mandatory	First Name
other_names	CharField	Nullable	Other Names
sex	CharField	Mandatory	Sex
title	CharField	Mandatory	Title
date_of_birth	DateField	Mandatory	Date Of Birth
national_id	CharField	Mandatory	National Id
marital_status	CharField	Mandatory	Marital Status
citizenship	CharField	Mandatory, Has Choices	Citizenship
telephone	CharField	Mandatory	Telephone
contact_address	TextField	Mandatory	Contact Address
email	CharField	Mandatory	Email
is_war_veteran	BooleanField	Mandatory	Is War Veteran

war_vet_number	CharField	Nullable	War Vet Number
is_ex_detainee	BooleanField	Mandatory	Is Ex Detainee
is_collaborator	BooleanField	Mandatory	Is Collaborator
is_uniformed_forces	BooleanField	Mandatory	Is Uniformed Forces
force_number	CharField	Nullable	Force Number
is_business_person	BooleanField	Mandatory	Is Business Person
is_youth	BooleanField	Mandatory	Is Youth
is_civil_servant	BooleanField	Mandatory	Is Civil Servant
ec_number	CharField	Nullable	Ec Number
civil_service_years	PositiveIntegerField	Nullable	Civil Service Years
is_mp_senator	BooleanField	Mandatory	Is Mp Senator
is_disabled	BooleanField	Mandatory	Is Disabled
is_zepdra	BooleanField	Mandatory	Is Zepdra
is_councilor	BooleanField	Mandatory	Is Councilor
is_traditional_chief	BooleanField	Mandatory	Is Traditional Chief
is_ordinary	BooleanField	Mandatory	Is Ordinary
is_other	BooleanField	Mandatory	Is Other
other_category_specify	CharField	Nullable	Other Category Specify
years_of_service	IntegerField	Nullable	Years Of Service
mfa_email_otp	CharField	Nullable	Mfa Email Otp
mfa_otp_created_at	DateTimeField	Nullable	Mfa Otp Created At
mfa_verified_for_session	BooleanField	Mandatory	Mfa Verified For Session
mfa_otp_failed_attempts	IntegerField	Mandatory	Mfa Otp Failed Attempts

Model: StaffProfile (Table: onlineapplicationapi_staffprofile)

Description: StaffProfile(id, user, role, department, channels, privileges, national_id, phone_number, mfa_email_otp, mfa_otp_created_at, mfa_verified_for_session)

Table A-StaffProfile: Data Dictionary for StaffProfile (11 Fields)

Field Name	Field Data Type	Constraints / Choices	Verbose Label / Purpose
id	BigAutoField	Mandatory	Id
user	OneToOneField	Mandatory	User
role	CharField	Mandatory, Has Choices	Role
department	CharField	Mandatory	Department
channels	JSONField	Mandatory	Channels
privileges	JSONField	Mandatory	Privileges
national_id	CharField	Nullable	National Id
phone_number	CharField	Nullable	Phone Number
mfa_email_otp	CharField	Nullable	Mfa Email Otp
mfa_otp_created_at	DateTimeField	Nullable	Mfa Otp Created At
mfa_verified_for_session	BooleanField	Mandatory	Mfa Verified For Session

Model: TitleDeedApplication (Table: onlineapplicationapi_titledeedapplication)

Description: TitleDeedApplication(id, reference_number, farmer, is_assisted_capture, assisted_by, walk_in_citizen_name, walk_in_national_id, walk_in_phone, province, district, old_district, farm_id, farm_name, farm_extent, arable_area, tenure_document_type, nearest_town, distance_to_town, water_source_type, water_source_distance, status, current_stage_entered_at, current_stage_due_date, is_overdue, resettlement_offer_letter_verified, resettlement_spouse_cleared, resettlement_estate_cleared, resettlement_verified_at, cadastral_id, statutory_fee_receipt, statutory_fee_paid, lims_verification_status, lims_verified_at, has_disputes, dispute_nature, dispute_comments, zlc_verification_status, zlc_cleared_at, boundary_coordinates, spatial_overlap_ratio, overlapping_parcels_count, overlapping_parcels_details, gis_validated_at, ai_conflict_score, ai_conflict_level, ai_conflict_details, survey_farm_name_confirmed, survey_hectarage_confirmed, survey_fee, is_survey_fee_paid, is_surveyed, survey_approved_at, survey_diagram, improvements_description, improvements_value, base_land_value, base_rate_per_hectare, valuation_amount, valuation_approved, valuated_at, valuation_report, director_approved, director_signed_at, director_comments, deed_registration_number,

deed_seal_date, deed_document, deed_issued_at, is_cash_payment, purchase_price, calculated_land_value, manual_valuation_price, is_manually_valuated, land_class_discount, water_premium, town_premium, ecological_region, payment_plan, selected_bank, payment_method_selected_at, payment_verified, application_fee_paid, collection_notification_sent, rejection_reason, queried_by, appeal_reason, tenure_document, national_id_copy, other_attachments, war_vet_id_copy, employment_proof, subdivision_number, plot_number, farm_survey_status, gazette_date, distance_to_main_road, natural_region, selected_land_classes, selected_water_sources, water_capacity_details, improvements_list, farmer_categories_list, beneficiary_photo_front, beneficiary_photo_side, beneficiary_id_front, beneficiary_id_back, beneficiary_signature, beneficiary_signature_date, police_clearance, tax_clearance, official_compiled_by, official_checked_by, idempotency_key, ai_risk_reasons, ai_model_version, ai_assessed_at, tracking_pin_hash, tracking_pin_sent_at, tracking_pin_failed_attempts, tracking_pin_locked_until, officer_comments, created_at, updated_at)

Table A-TitleDeedApplication: Data Dictionary for TitleDeedApplication (123 Fields)

Field Name	Field Data Type	Constraints / Choices	Verbose Label / Purpose
id	BigAutoField	Mandatory	Id
reference_number	CharField	Nullable	Reference Number
farmer	OneToOneField	Nullable	Farmer
is_assisted_capture	BooleanField	Mandatory	Is Assisted Capture
assisted_by	ForeignKey	Nullable	Assisted By
walk_in_citizen_name	CharField	Nullable	Walk In Citizen Name
walk_in_national_id	CharField	Nullable	Walk In National Id
walk_in_phone	CharField	Nullable	Walk In Phone
province	CharField	Nullable	Province
district	CharField	Mandatory	District
old_district	CharField	Nullable	Old District
farm_id	CharField	Nullable	Farm Id
farm_name	CharField	Mandatory	Farm Name
farm_extent	DecimalField	Mandatory	Farm Extent
arable_area	DecimalField	Mandatory	Arable Area

tenure_document_type	CharField	Nullable	Tenure Document Type
nearest_town	CharField	Nullable	Nearest Town
distance_to_town	DecimalField	Mandatory	Distance To Town
water_source_type	JSONField	Mandatory	Water Source Type
water_source_distance	CharField	Nullable	Water Source Distance
status	CharField	Mandatory, Has Choices	Status
current_stage_entered_at	DateTimeField	Mandatory	Current Stage Entered At
current_stage_due_date	DateTimeField	Nullable	Current Stage Due Date
is_overdue	BooleanField	Mandatory	Is Overdue
resettlement_offer_letter_verified	BooleanField	Mandatory	Resettlement Offer Letter Verified
resettlement_spouse_cleared	BooleanField	Mandatory	Resettlement Spouse Cleared
resettlement_estate_cleared	BooleanField	Mandatory	Resettlement Estate Cleared
resettlement_verified_at	DateTimeField	Nullable	Resettlement Verified At
cadastral_id	CharField	Nullable	Cadastral Id
statutory_fee_receipt	CharField	Nullable	Statutory Fee Receipt
statutory_fee_paid	BooleanField	Mandatory	Statutory Fee Paid
lims_verification_status	BooleanField	Mandatory	Lims Verification Status
lims_verified_at	DateTimeField	Nullable	Lims Verified At
has_disputes	BooleanField	Mandatory	Has Disputes
dispute_nature	CharField	Nullable	Dispute Nature
dispute_comments	TextField	Nullable	Dispute Comments
zlc_verification_status	BooleanField	Mandatory	Zlc Verification Status

zlc_cleared_at	DateTimeField	Nullable	Zlc Cleared At
boundary_coordinates	JSONField	Nullable	Boundary Coordinates
spatial_overlap_ratio	FloatField	Mandatory	Spatial Overlap Ratio
overlapping_parcels_count	IntegerField	Mandatory	Overlapping Parcels Count
overlapping_parcels_details	JSONField	Mandatory	Overlapping Parcels Details
gis_validated_at	DateTimeField	Nullable	Gis Validated At
ai_conflict_score	IntegerField	Mandatory	Ai Conflict Score
ai_conflict_level	CharField	Mandatory, Has Choices	Ai Conflict Level
ai_conflict_details	JSONField	Mandatory	Ai Conflict Details
survey_farm_name_confirmed	CharField	Nullable	Survey Farm Name Confirmed
survey_hectarage_confirmed	DecimalField	Nullable	Survey Hectarage Confirmed
survey_fee	DecimalField	Mandatory	Survey Fee
is_survey_fee_paid	BooleanField	Mandatory	Is Survey Fee Paid
is_surveyed	BooleanField	Mandatory	Is Surveyed
survey_approved_at	DateTimeField	Nullable	Survey Approved At
survey_diagram	FileField	Nullable	Survey Diagram
improvements_description	TextField	Nullable	Improvements Description
improvements_value	DecimalField	Mandatory	Improvements Value
base_land_value	DecimalField	Mandatory	Base Land Value
base_rate_per_hectare	DecimalField	Mandatory	Base Rate Per Hectare
valuation_amount	DecimalField	Mandatory	Valuation Amount
valuation_approved	BooleanField	Mandatory	Valuation Approved

valuated_at	DateTimeField	Nullable	Valuated At
valuation_report	FileField	Nullable	Valuation Report
director_approved	BooleanField	Mandatory	Director Approved
director_signed_at	DateTimeField	Nullable	Director Signed At
director_comments	TextField	Nullable	Director Comments
deed_registration_number	CharField	Nullable	Deed Registration Number
deed_seal_date	DateField	Nullable	Deed Seal Date
deed_document	FileField	Nullable	Deed Document
deed_issued_at	DateTimeField	Nullable	Deed Issued At
is_cash_payment	BooleanField	Mandatory	Is Cash Payment
purchase_price	DecimalField	Mandatory	Purchase Price
calculated_land_value	DecimalField	Mandatory	Calculated Land Value
manual_valuation_price	DecimalField	Mandatory	Manual Valuation Price
is_manually_valuated	BooleanField	Mandatory	Is Manually Valuated
land_class_discount	DecimalField	Mandatory	Land Class Discount
water_premium	DecimalField	Mandatory	Water Premium
town_premium	DecimalField	Mandatory	Town Premium
ecological_region	CharField	Nullable	Ecological Region
payment_plan	CharField	Nullable	Payment Plan
selected_bank	CharField	Nullable	Selected Bank
payment_method_selected_at	DateTimeField	Nullable	Payment Method Selected At
payment_verified	BooleanField	Mandatory	Payment Verified
application_fee_paid	BooleanField	Mandatory	Application Fee Paid

collection_notification_sent	BooleanField	Mandatory	Collection Notification Sent
rejection_reason	TextField	Nullable	Rejection Reason
queried_by	CharField	Nullable	Queried By
appeal_reason	TextField	Nullable	Appeal Reason
tenure_document	FileField	Nullable	Tenure Document
national_id_copy	FileField	Nullable	National Id Copy
other_attachments	FileField	Nullable	Other Attachments
war_vet_id_copy	FileField	Nullable	War Vet Id Copy
employment_proof	FileField	Nullable	Employment Proof
subdivision_number	CharField	Nullable	Subdivision Number
plot_number	CharField	Nullable	Plot Number
farm_survey_status	CharField	Nullable	Farm Survey Status
gazette_date	DateField	Nullable	Gazette Date
distance_to_main_road	DecimalField	Mandatory	Distance To Main Road
natural_region	CharField	Nullable	Natural Region
selected_land_classes	JSONField	Mandatory	Selected Land Classes
selected_water_sources	JSONField	Mandatory	Selected Water Sources
water_capacity_details	TextField	Nullable	Water Capacity Details
improvements_list	JSONField	Mandatory	Improvements List
farmer_categories_list	JSONField	Mandatory	Farmer Categories List
beneficiary_photo_front	FileField	Nullable	Beneficiary Photo Front
beneficiary_photo_side	FileField	Nullable	Beneficiary Photo Side
beneficiary_id_front	FileField	Nullable	Beneficiary Id Front

beneficiary_id_back	FileField	Nullable	Beneficiary Id Back
beneficiary_signature	FileField	Nullable	Beneficiary Signature
beneficiary_signature_date	DateField	Nullable	Beneficiary Signature Date
police_clearance	FileField	Nullable	Police Clearance
tax_clearance	FileField	Nullable	Tax Clearance
official_compiled_by	CharField	Nullable	Official Compiled By
official_checked_by	CharField	Nullable	Official Checked By
idempotency_key	CharField	Nullable	Idempotency Key
ai_risk_reasons	JSONField	Mandatory	Ai Risk Reasons
ai_model_version	CharField	Mandatory	Ai Model Version
ai_assessed_at	DateTimeField	Nullable	Ai Assessed At
tracking_pin_hash	CharField	Nullable	Tracking Pin Hash
tracking_pin_sent_at	DateTimeField	Nullable	Tracking Pin Sent At
tracking_pin_failed_attempts	IntegerField	Mandatory	Tracking Pin Failed Attempts
tracking_pin_locked_until	DateTimeField	Nullable	Tracking Pin Locked Until
officer_comments	TextField	Nullable	Officer Comments
created_at	DateTimeField	Mandatory	Created At
updated_at	DateTimeField	Mandatory	Updated At

Model: SpouseDetail (Table: onlineapplicationapi_spousedetail)

Description: SpouseDetail(id, application, surname, first_name, national_id, citizenship, is_war_veteran, war_vet_id_copy, is_ex_detainee, is_collaborator, is_civil_servant, civil_service_years, employment_proof, title, other_names, date_of_birth, telephone, marriage_type, marriage_certificate_number, photo_front, photo_side, id_front, id_back, marriage_certificate, signature, signature_date, created_at, updated_at)

Table A-SpouseDetail: Data Dictionary for SpouseDetail (28 Fields)

Field Name	Field Data Type	Constraints / Choices	Verbose Label / Purpose
id	BigAutoField	Mandatory	Id
application	ForeignKey	Mandatory	Application
surname	CharField	Nullable	Surname
first_name	CharField	Nullable	First Name
national_id	CharField	Nullable	National Id
citizenship	CharField	Mandatory	Citizenship
is_war_veteran	BooleanField	Mandatory	Is War Veteran
war_vet_id_copy	FileField	Nullable	War Vet Id Copy
is_ex_detainee	BooleanField	Mandatory	Is Ex Detainee
is_collaborator	BooleanField	Mandatory	Is Collaborator
is_civil_servant	BooleanField	Mandatory	Is Civil Servant
civil_service_years	PositiveIntegerField	Nullable	Civil Service Years
employment_proof	FileField	Nullable	Employment Proof
title	CharField	Nullable	Title
other_names	CharField	Nullable	Other Names
date_of_birth	DateField	Nullable	Date Of Birth
telephone	CharField	Nullable	Telephone
marriage_type	CharField	Nullable	Marriage Type
marriage_certificate_number	CharField	Nullable	Marriage Certificate Number
photo_front	FileField	Nullable	Photo Front
photo_side	FileField	Nullable	Photo Side

id_front	FileField	Nullable	Id Front
id_back	FileField	Nullable	Id Back
marriage_certificate	FileField	Nullable	Marriage Certificate
signature	FileField	Nullable	Signature
signature_date	DateField	Nullable	Signature Date
created_at	DateTimeField	Mandatory	Created At
updated_at	DateTimeField	Mandatory	Updated At

Model: WorkflowTransition (Table: onlineapplicationapi_workflowtransition)

Description: Database-driven workflow transitions defining sequential stages, allowed role, return capabilities, and configurable SLAs.

Table A-WorkflowTransition: Data Dictionary for WorkflowTransition (8 Fields)

Field Name	Field Data Type	Constraints / Choices	Verbose Label / Purpose
id	BigAutoField	Mandatory	Id
from_stage	CharField	Mandatory, Has Choices	From Stage
to_stage	CharField	Mandatory, Has Choices	To Stage
allowed_role	CharField	Mandatory, Has Choices	Allowed Role
is_return	BooleanField	Mandatory	Is Return
requires_comment	BooleanField	Mandatory	Requires Comment
description	CharField	Mandatory	Description
sla_days	PositiveIntegerField	Mandatory	Sla Days

Model: StageSLA (Table: onlineapplicationapi_stagesla)

Description: Configurable Stage SLAs with notification recipients.

Table A-StageSLA: Data Dictionary for StageSLA (5 Fields)

Field Name	Field Data Type	Constraints / Choices	Verbose Label / Purpose
id	BigAutoField	Mandatory	Id
stage	CharField	Mandatory, Has Choices	Stage
department_name	CharField	Mandatory	Department Name
sla_days	PositiveIntegerField	Mandatory	Sla Days
notification_email	CharField	Mandatory	Notification Email

Model: SurveyTariff (Table: onlineapplicationapi_surveytariff)

Description: Configurable Survey Tariff table based on parcel area in hectares.

Table A-SurveyTariff: Data Dictionary for SurveyTariff (6 Fields)

Field Name	Field Data Type	Constraints / Choices	Verbose Label / Purpose
id	BigAutoField	Mandatory	Id
min_area_ha	DecimalField	Mandatory	Min Area Ha
max_area_ha	DecimalField	Mandatory	Max Area Ha
base_fee_usd	DecimalField	Mandatory	Base Fee Usd
per_hectare_rate_usd	DecimalField	Mandatory	Per Hectare Rate Usd
description	CharField	Mandatory	Description

Model: ValuationTariff (Table: onlineapplicationapi_valuationtariff)

Description: Configurable Valuation Tariff table: base rate per hectare by ecological region and land class.

Table A-ValuationTariff: Data Dictionary for ValuationTariff (5 Fields)

Field Name	Field Data Type	Constraints / Choices	Verbose Label / Purpose
id	BigAutoField	Mandatory	Id
region	CharField	Mandatory	Region

land_class	CharField	Mandatory	Land Class
base_rate_per_ha_usd	DecimalField	Mandatory	Base Rate Per Ha Usd
description	CharField	Mandatory	Description

Model: ApplicationAction (Table: onlineapplicationapi_applicationaction)

Description: Blockchain-style (hash-chained), append-only immutable audit ledger. Block modifications (UPDATE and DELETE) are prohibited at the model level.

Table A-ApplicationAction: Data Dictionary for ApplicationAction (10 Fields)

Field Name	Field Data Type	Constraints / Choices	Verbose Label / Purpose
id	BigAutoField	Mandatory	Id
application	ForeignKey	Mandatory	Application
officer	ForeignKey	Mandatory	Officer
from_stage	CharField	Mandatory	From Stage
to_stage	CharField	Mandatory	To Stage
action_type	CharField	Mandatory	Action Type
comment	TextField	Mandatory	Comment
previous_hash	CharField	Mandatory	Previous Hash
block_hash	CharField	Mandatory	Block Hash
created_at	DateTimeField	Mandatory	Created At

Model: EmailLog (Table: onlineapplicationapi_emaillog)

Description: EmailLog(id, recipient, subject, body, status, application, error_message, sent_at, attempts)

Table A-EmailLog: Data Dictionary for EmailLog (9 Fields)

Field Name	Field Data Type	Constraints / Choices	Verbose Label / Purpose

id	BigAutoField	Mandatory	Id
recipient	CharField	Mandatory	Recipient
subject	CharField	Mandatory	Subject
body	TextField	Mandatory	Body
status	CharField	Mandatory, Has Choices	Status
application	ForeignKey	Nullable	Application
error_message	TextField	Nullable	Error Message
sent_at	DateTimeField	Mandatory	Sent At
attempts	IntegerField	Mandatory	Attempts

Model: Payment (Table: onlineapplicationapi_payment)

Description: Server-governed payment record for the statutory processing fee (USD $10.00). Receipt numbers are generated and validated on the server only.

Table A-Payment: Data Dictionary for Payment (14 Fields)

Field Name	Field Data Type	Constraints / Choices	Verbose Label / Purpose
id	BigAutoField	Mandatory	Id
receipt_number	CharField	Mandatory	Receipt Number
amount	DecimalField	Mandatory	Amount
currency	CharField	Mandatory	Currency
provider	CharField	Mandatory	Provider
status	CharField	Mandatory	Status
beneficiary_name	CharField	Mandatory	Beneficiary Name
national_id	CharField	Mandatory	National Id
phone_number	CharField	Mandatory	Phone Number

is_used	BooleanField	Mandatory	Is Used
used_for_application	ForeignKey	Nullable	Used For Application
transaction_ref	CharField	Mandatory	Transaction Ref
created_at	DateTimeField	Mandatory	Created At
updated_at	DateTimeField	Mandatory	Updated At

Appendix B: Complete API Endpoint Inventory

Router Inventory: The following inventory documents key REST API endpoints registered within the Django URL router:

Table B-1: Complete Registered API Endpoint Inventory

URL Route Pattern	Route Name	Implementing Controller / Callback
api/v1/login/	login	onlineapplicationapi.views.LoginView
api/v1/logout/	logout	onlineapplicationapi.views.LogoutView
api/v1/check-session/	check-session	CheckSessionView
api/v1/user/profile/	user-profile	UserProfileView
api/v1/csrf-cookie/	csrf-cookie	SetCSRFCookieView
api/v1/password-reset/	password-reset	PasswordResetRequestView
api/v1/password-reset-confirm/	password-reset-confirm	PasswordResetConfirmView
api/v1/track/	track-application	TrackApplicationView
api/v1/gis/farms/	gis-farms	onlineapplicationapi.views.GISFarmsView
api/v1/applications/<int:pk>/ai-dispute-prediction/	ai-dispute-prediction	AIDisputePredictionView
api/v1/applications/<int:pk>/verify-audit-chain/	verify-audit-chain	BlockchainAuditVerifyView
api/v1/ai/what-if/	ai-what-if	onlineapplicationapi.views.AIWhatIfView
api/v1/ai/district-risk/	ai-district-risk	AIDistrictRiskView

api/v1/applications/<int:application_id>/document/<str:field_name>/	secure-document-download	SecureDocumentDownloadView
api/v1/^profiles/$	profile-list	FarmerProfileViewSet
api/v1/^profiles\.(?P<format>[a-z0-9]+)/?$	profile-list	FarmerProfileViewSet
api/v1/^profiles/(?P<pk>[^/.]+)/$	profile-detail	FarmerProfileViewSet
api/v1/^profiles/(?P<pk>[^/.]+)\.(?P<format>[a-z0-9]+)/?$	profile-detail	FarmerProfileViewSet
api/v1/^applications/$	application-list	TitleDeedApplicationViewSet
api/v1/^applications\.(?P<format>[a-z0-9]+)/?$	application-list	TitleDeedApplicationViewSet
api/v1/^applications/assisted-capture/$	application-assisted-capture	TitleDeedApplicationViewSet
api/v1/^applications/assisted-capture\.(?P<format>[a-z0-9]+)/?$	application-assisted-capture	TitleDeedApplicationViewSet
api/v1/^applications/calculate-pricing/$	application-calculate-pricing	TitleDeedApplicationViewSet
api/v1/^applications/calculate-pricing\.(?P<format>[a-z0-9]+)/?$	application-calculate-pricing	TitleDeedApplicationViewSet
api/v1/^applications/dashboard_stats/$	application-dashboard-stats	TitleDeedApplicationViewSet
api/v1/^applications/dashboard_stats\.(?P<format>[a-z0-9]+)/?$	application-dashboard-stats	TitleDeedApplicationViewSet

api/v1/^applications/stats/$	application-stats	TitleDeedApplicationViewSet
api/v1/^applications/stats\.(?P<format>[a-z0-9]+)/?$	application-stats	TitleDeedApplicationViewSet
api/v1/^applications/(?P<pk>[^/.]+)/$	application-detail	TitleDeedApplicationViewSet
api/v1/^applications/(?P<pk>[^/.]+)\.(?P<format>[a-z0-9]+)/?$	application-detail	TitleDeedApplicationViewSet
api/v1/^applications/(?P<pk>[^/.]+)/appeal_rejection/$	application-appeal-rejection	TitleDeedApplicationViewSet
api/v1/^applications/(?P<pk>[^/.]+)/appeal_rejection\.(?P<format>[a-z0-9]+)/?$	application-appeal-rejection	TitleDeedApplicationViewSet
api/v1/^applications/(?P<pk>[^/.]+)/approve/$	application-approve	TitleDeedApplicationViewSet
api/v1/^applications/(?P<pk>[^/.]+)/approve\.(?P<format>[a-z0-9]+)/?$	application-approve	TitleDeedApplicationViewSet
api/v1/^applications/(?P<pk>[^/.]+)/conclude-application/$	application-conclude-application	TitleDeedApplicationViewSet
api/v1/^applications/(?P<pk>[^/.]+)/conclude-application\.(?P<format>[a-z0-9]+)/?$	application-conclude-application	TitleDeedApplicationViewSet
api/v1/^applications/(?P<pk>[^/.]+)/escalate/$	application-escalate	TitleDeedApplicationViewSet
api/v1/^applications/(?P<pk>[^/.]+)/escalate\.(?P<format>[a-z0-9]+)/?$	application-escalate	TitleDeedApplicationViewSet
api/v1/^applications/(?P<pk>[^/.]+)/final_approve/$	application-final-approve	TitleDeedApplicationViewSet
api/v1/^applications/(?P<pk>[^/.]+)/final_approve\.(?P<format>[a-	application-final-	TitleDeedApplicationViewSet

z0-9]+)/?$	approve	
api/v1/^applications/(?P<pk>[^/.]+)/generate_agreement/$	application-generate-agreement	TitleDeedApplicationViewSet
api/v1/^applications/(?P<pk>[^/.]+)/generate_agreement\.(?P<format>[a-z0-9]+)/?$	application-generate-agreement	TitleDeedApplicationViewSet
api/v1/^applications/(?P<pk>[^/.]+)/mark_surveyed/$	application-mark-surveyed	TitleDeedApplicationViewSet
api/v1/^applications/(?P<pk>[^/.]+)/mark_surveyed\.(?P<format>[a-z0-9]+)/?$	application-mark-surveyed	TitleDeedApplicationViewSet
api/v1/^applications/(?P<pk>[^/.]+)/pay_survey_fee/$	application-pay-survey-fee	TitleDeedApplicationViewSet

Appendix C: Sample System Artifacts

Synthetic Artifacts: The following artifacts illustrate system outputs using synthetic demonstration data. No personal data appears in these records:

C.1 Sample Digital Statutory Processing Fee Receipt

```
========================================================================
        MINISTRY OF LANDS, AGRICULTURE, FISHERIES, WATER & RURAL DEV
                  TITLE DEEDS ONE STOP CENTER: OFFICIAL RECEIPT
========================================================================
Receipt Number   : REC-2026-88492-ZW
Date of Payment  : 27 October 2026 09:14:22 CAT
Payer Full Name  : MOYO, NYASHA (Sample Beneficiary)
National ID No.  : 63-8829104-B-42
Payment Channel  : EcoCash Mobile Wallet (Simulator)
Transaction Ref  : EC-99482018471
Statutory Fee    : Title Deed Processing Fee (Paper Form Step 0)
Amount Paid      : $10.00 USD
Payment Status   : CONFIRMED (Statutory Unlock Code: UNL-99481)
========================================================================
This digital receipt confirms valid payment under Statutory Instrument 142.
Please retain this receipt code to unlock and submit your digital dossier.
========================================================================
```

C.2 Sample Statutory Submission Confirmation

```
========================================================================
             STATUTORY APPLICATION SUBMISSION CONFIRMATION
========================================================================
Application Reference  : TD-2026-MSH-00412
Applicant Full Name    : MOYO, NYASHA
Spouse Co-Applicant    : MOYO, TARIRO (Joint Registration Confirmed)
Target Land Parcel     : Farm 14, Subdivision B, Marondera District
Allocated Hectares     : 45.5000 Hectares (Agro-Ecological Region IIa)
Current Status         : PENDING (Assigned to Resettlement Verification)
Date of Submission     : 27 October 2026 10:28:45 CAT
Audit Block Hash       : 4a8f9c1b3e7d2a5e8f0a1c3b5d7e9f2a4c6e8b0d2f4a6c8e0b2d4f6a8c0e2b4d
========================================================================
```

C.3 Sample Cryptographic SHA-256 Audit Ledger Chain

```
BLOCK 0 (GENESIS ANCHOR):
  Application ID : TD-2026-MSH-00412
  Stage          : INTAKE_INITIALIZED
  Previous Hash  : 0000000000000000000000000000000000000000000000000000000000000000
  Timestamp      : 2026-10-27T10:28:45.102Z
  Block Hash     : 4a8f9c1b3e7d2a5e8f0a1c3b5d7e9f2a4c6e8b0d2f4a6c8e0b2d4f6a8c0e2b4d

BLOCK 1 (STAGE 1: RESETTLEMENT VERIFICATION):
  Application ID : TD-2026-MSH-00412
  Stage          : RESETTLEMENT_VERIFIED
  Officer ID     : staff_officer_resettlement_01 (User ID: 14)
  Action Taken   : APPROVE (Offer letter inspected & verified on-site)
  Previous Hash  : 4a8f9c1b3e7d2a5e8f0a1c3b5d7e9f2a4c6e8b0d2f4a6c8e0b2d4f6a8c0e2b4d
  Timestamp      : 2026-10-27T14:12:08.441Z
```

```
  Block Hash     : 8b2c4e6a0d2f4a6c8e0b2d4f6a8c0e2b4d4a8f9c1b3e7d2a5e8f0a1c3b5d7e9f
```

Appendix D: Automated Test Output Evidence

Master Test Suite Output: The following verbatim console output records the empirical execution of the aligned statutory test suite:

```
$ python manage.py test onlineapplicationapi.test_aligned_suite
Creating test database for alias 'default'...
Found 15 test(s).
System check identified no issues (0 silenced).
test_01_farmer_registration
(onlineapplicationapi.test_aligned_suite.AlignedStatutoryWorkflowTests) ... ok
test_02_staff_mfa_login
(onlineapplicationapi.test_aligned_suite.AlignedStatutoryWorkflowTests) ... ok
test_03_payment_gate_processing
(onlineapplicationapi.test_aligned_suite.AlignedStatutoryWorkflowTests) ... ok
test_04_receipt_generation_code
(onlineapplicationapi.test_aligned_suite.AlignedStatutoryWorkflowTests) ... ok
test_05_application_intake_full_fields
(onlineapplicationapi.test_aligned_suite.AlignedStatutoryWorkflowTests) ... ok
test_06_stage1_resettlement_approval
(onlineapplicationapi.test_aligned_suite.AlignedStatutoryWorkflowTests) ... ok
test_07_zero_privilege_lockout
(onlineapplicationapi.test_aligned_suite.AlignedStatutoryWorkflowTests) ... ok
test_08_stage2_and_3_lims_zlc_clearance
(onlineapplicationapi.test_aligned_suite.AlignedStatutoryWorkflowTests) ... ok
test_09_stage4_gis_boundary_validation
(onlineapplicationapi.test_aligned_suite.AlignedStatutoryWorkflowTests) ... ok
test_10_stage5_surveyor_general_approval
(onlineapplicationapi.test_aligned_suite.AlignedStatutoryWorkflowTests) ... ok
test_11_stage6_compounded_valuation
(onlineapplicationapi.test_aligned_suite.AlignedStatutoryWorkflowTests) ... ok
test_12_stage7_director_executive_approval
(onlineapplicationapi.test_aligned_suite.AlignedStatutoryWorkflowTests) ... ok
test_13_stage8_deeds_registry_issuance
(onlineapplicationapi.test_aligned_suite.AlignedStatutoryWorkflowTests) ... ok
test_14_ready_for_collection_state
(onlineapplicationapi.test_aligned_suite.AlignedStatutoryWorkflowTests) ... ok
test_15_sha256_audit_chain_integrity
(onlineapplicationapi.test_aligned_suite.AlignedStatutoryWorkflowTests) ... ok
----------------------------------------------------------------------
Ran 15 tests in 57.072s

OK
Destroying test database for alias 'default'...
```

Appendix E: Codebase Directory Tree and Architecture Manifest

Repository Layout: The hierarchical file structure of the `onestopproject` repository is detailed below with functional descriptions:

```
onestopproject/
├── TitleDeeds.apk                    # Compiled release Android APK (9.67 MB, signed v2/v3)
├── build_apk.ps1                     # Automated build script for Android release packaging
├── documentation/                    # Technical project documentation artifacts
│   ├── README.md                     # Documentation directory overview and build
instructions
│   ├── inventory.md                  # Automated codebase inventory manifest
│   ├── models_data.json              # Extracted schema for 11 Django models (257 fields)
│   ├── routes_data.json              # Extracted API router inventory (305 endpoints)
│   ├── diagrams/                     # 16 standalone PNG vector system diagrams
│   └── screenshots/                  # 18 real unedited application screenshots
└── onlineportal/                     # Core application source directory
    ├── onlineapplication/            # Django 5.2 backend project
    │   ├── manage.py                 # Django administrative command CLI
    │   ├── db.sqlite3                # Local SQLite database (50,000 synthetic records)
    │   ├── requirements.txt          # Python dependency specification
    │   ├── onlineapplication/        # Project configuration (settings, urls, wsgi, asgi)
    │   └── onlineapplicationapi/     # Main application package
    │       ├── models.py             # 11 relational database models
    │       ├── views.py              # REST API viewsets, authentication, and transitions
    │       ├── serializers.py        # DRF serializers for request/response payloads
    │       ├── pricing.py            # Compounded land valuation and pricing engine
    │       ├── test_aligned_suite.py  # Master statutory 15-test automated suite (100% OK)
    │       ├── ml/                   # AI dispute predictor package
    │       │   ├── dispute_predictor.py            # Scikit-learn inference engine
    │       │   └── synthetic_dispute_training_data.csv # 1,000 synthetic training records
    │       └── management/commands/   # Custom management commands
    │           ├── verify_audit_ledger.py          # SHA-256 chain integrity validator
    │           └── init_tariffs.py                 # Tariff seeding script
    └── onlineportal-ui/              # React 18 frontend and mobile project
        ├── package.json              # Node.js dependencies and build scripts
        ├── vite.config.js            # Vite development server and bundling configuration
        ├── capacitor.config.json     # Capacitor 6 Android mobile configuration
        ├── android/                  # Native Android container project
        └── src/                      # Frontend source code
            ├── components/           # Reusable UI components (SignaturePad, PhotoCapture)
            ├── pages/                # Full application views (ApplicationForm,
OfficerDashboard)
            └── services/             # API communication layer (api.js, auth.js)
```

Appendix F: Change Log of Post-Baseline Defect Fixes

Resolution Log: This log chronicles technical issues identified and resolved after the initial baseline build:

- Reports and Audit Logs Navigation Fix: Added explicit routing guards in `OfficerDashboard.jsx` allowing non-admin departmental officers read-only inspection of departmental SLA reports.
- Citizen Account Creation CSRF Fix: Resolved a form submission failure on `/api/auth/register/` by decorating `RegisterView` with `@permission_classes([AllowAny])` to permit unauthenticated citizen onboarding.
- Officer Dashboard APK Banner Removal: Eliminated an unwanted mobile APK download banner from internal departmental screens by adding role-reactive JSX conditional guards.
- User Management Role Auto-Fill: Updated the administrative role assignment modal in `admin-portal` to automatically pre-select the staff member's current role rather than defaulting to empty.
- Full-Screen Touch Signature Pad: Refactored `SignaturePad.jsx` with responsive aspect-ratio canvas scaling to prevent signature clipping on mobile screens.
- Standardized 'Label: value' Formatting: Standardized UI readouts across citizen dashboards and officer review modals to enforce clean 'Label: value' typographic formatting.

Appendix G: Formal References & Bibliography

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